

RTL Design Sherpa

STREAM Subsystem Hardware Architecture Specification 0.90

January 3, 2026

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2

Document Information

2.1

STREAM Hardware Architecture Specification

Document Number: STREAM-HAS-001 **Version:** 0.95 **Status:** Draft **Classification:** Open Source - Apache 2.0 License

2.2

Document Purpose

This Hardware Architecture Specification (HAS) provides a high-level architectural overview of the STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory) subsystem. It describes the system-level design, external interfaces, performance characteristics, and integration requirements without detailing internal implementation specifics.

Target Audience: - System architects evaluating STREAM for integration - Hardware engineers planning system-level integration - Software engineers developing drivers and firmware - Verification engineers planning system-level testing

Companion Documents: - STREAM Micro-Architecture Specification (MAS) - Detailed block-level implementation - STREAM Product Requirements Document (PRD) - Requirements and rationale

2.3

References

Reference Documents

ID	Document	Description
[REF-1]	STREAM MAS v0.90	Micro-Architecture Specification
[REF-2]	STREAM PRD	Product

ID	Document	Description
		Requirements Document
[REF-3]	ARM AMBA AXI4	AXI4 Protocol Specification
[REF-4]	ARM AMBA APB	APB Protocol Specification

2.4 Terminology

Terminology and Definitions

Term	Definition
AXI	Advanced eXtensible Interface - ARM AMBA high-performance bus
APB	Advanced Peripheral Bus - ARM AMBA low-power configuration bus
Beat	Single data transfer on AXI bus (one clock cycle of valid data)
Burst	Sequence of consecutive beats forming a single AXI transaction
Channel	Independent DMA transfer context (STREAM supports 8 channels)
Descriptor	256-bit data structure defining a single DMA transfer operation
DMA	Direct Memory Access - data transfer without CPU involvement
HAS	Hardware Architecture Specification
MAS	Micro-Architecture Specification
MonBus	Monitor Bus - internal debug/trace event streaming interface
Scatter-Gather	DMA mode using linked descriptors for non-contiguous transfers

2.5 Revision History

Document Revision History

Version	Date	Author	Description
0.90	2026-01-03	seang	Initial HAS release
0.91	2026-05-14	seang	Sync to RTL changes since 2026-04-17 (commit be4e5a91); regenerate PDFs/DOCX.
0.92	2026-06-05	seang	Sync to RTL state at 2026-06-05 (17 commits since be4e5a91). Documents (1) the 64→128-bit monbus packet migration + new 64-bit side-band timestamp wire — referenced via docs/markdown/RTLAmba/includes/monitor_package_spec.md, with cfg_ts_append_* removed and m_axil records locked to 3 beats; (2) new APB channel-observation register set (OBS_CTRL / OBS_FLAGS / OBS_DATA0 / OBS_DATA1) that exposes scheduler error stickies + timeout per channel; (3) BOTH-end descriptor-path monitoring with distinct (UNIT_ID, AGENT_ID) for fetch-side vs. consume-side, under the new per-port + global SV-parameter monitor methodology; (4) three RTL fixes — drain-ctrl stale-view race + post-flop wvalid gate

Version	Date	Author	Description
			(a82627af), registered w_arb_request for 8-channel timing closure (4e8f9e02), reset on SRAM avail outputs (b619eee9); (5) stream_core_mon duplicate cleanup (7291c4ef).
0.93	2026-07-02	seang	Monitor registers relocated to a separate stream_mon_regs regfile at 0x1000 (single APB slave); APB address decode widened to 13 bits / 8 KB so the monitor block is addressable (ch04_interfaces/02_apb_slave.md). Companion MAS documents SCHED_TIMEOUT_LIMIT + the recoverable scheduler write-progress timeout and functional descriptor prefetch.
0.94	2026-07-07	seang	Documented the kick-burst fast-path interface (top-level i_kick_burst_mask / i_kick_burst_addr ports) that starts any subset of channels on a single clock cycle, bypassing the serial APB CHn_CTRL kick — added the interface signal table + semantics to ch04_interfaces/02_apb_slave.md. Reference

Version	Date	Author	Description
0.95	2026-07-09	seang	integration is the NexysA7 char harness (CH_KICK_ADDR shadow registers + KICK_GO bitmask CSR). Documented read-ahead descriptor prefetch: the scheduler read side advances to the next chained legacy descriptor while the write side drains the current one, eliminating the per-descriptor boundary bubble (cross-descriptor streaming). Added a section to ch05_performance/01_throughput.md with on-silicon A/B (Nexys A7, single bitstream: 95.3% datapath util prefetch-on vs 76.1% off at 64-beat descriptors; row-major EXT 99.7% util bubble-free; transpose single-beat latency-bound). Runtime SCHED_CONFIG.RD_PREFETCH_EN, default on.

Last Updated: 2026-07-02

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3 Purpose and Scope

3.1 Document Purpose

This Hardware Architecture Specification (HAS) defines the high-level architecture of the STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory) subsystem. It serves as the primary reference for:

- **System Integration** - Understanding STREAM's external interfaces and system-level requirements
- **Performance Planning** - Evaluating throughput, latency, and resource characteristics
- **Driver Development** - Programming model and software interface requirements
- **Verification Planning** - System-level test scenarios and coverage requirements

This document complements the STREAM Micro-Architecture Specification (MAS), which provides detailed block-level implementation specifics.

3.2 Scope

3.2.1 In Scope

This specification covers:

1. **System Architecture**
 - Block-level functional partitioning
 - Data flow through the subsystem
 - Multi-channel operation model
2. **External Interfaces**
 - APB configuration slave interface
 - AXI4 master interfaces (descriptor fetch, data read, data write)
 - Monitor bus (MonBus) output interface
3. **Performance Characteristics**
 - Throughput targets and limiting factors
 - Latency breakdown for typical operations
 - Resource estimates for FPGA/ASIC targets

4. Integration Requirements

- Clock and reset requirements
- System-level constraints
- Verification strategy overview

3.2.2 Out of Scope

This specification does not cover:

- **Detailed Micro-Architecture** - See STREAM MAS for block-level implementation
 - **RTL Implementation Details** - Register-level timing, pipeline stages
 - **Testbench Architecture** - See verification documentation
 - **Software Driver Implementation** - See software programming guide
-

3.3 Design Philosophy

3.3.1 Tutorial-Focused Simplicity

STREAM is intentionally designed as a beginner-friendly DMA engine tutorial. Key simplifications include:

Aspect	Simplification	Rationale
Address Alignment	Aligned addresses only	Eliminates complex fixup logic
Transfer Length	Specified in beats	Simplifies datapath math
Descriptor Chaining	Linear chains only	No circular buffer management
Flow Control	Simple transaction limits	No credit management complexity

These simplifications make STREAM an ideal learning platform while maintaining production-quality RTL practices.

3.3.2 Relationship to RAPIDS

STREAM is derived from the RAPIDS (Rapid AXI Programmable In-band Descriptor System) architecture but removes:

- Network interfaces (source/sink paths)
- Address alignment fixup logic
- Credit-based flow control
- Control read/write engines

This creates a pure memory-to-memory DMA engine suitable for understanding core DMA concepts.

3.4 Intended Audience

Audience	Primary Use
System Architects	Evaluate STREAM for system integration
Hardware Engineers	Plan physical integration and timing closure
Software Engineers	Develop drivers and firmware
Verification Engineers	Plan system-level test coverage

Last Updated: 2026-01-03

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4 Document Conventions

4.1 Notation Conventions

4.1.1 Signal Naming

Prefix/Suffix	Meaning	Example
i_	Input signal	i_clk, i_data
o_	Output signal	o_valid, o_ready
r_	Registered signal	r_state, r_count

Prefix/Suffix	Meaning	Example
w_	Combinational wire	w_next_state
_n	Active-low signal	aresetn, rst_n

4.1.2 Interface Prefixes

Prefix	Protocol	Direction
s_apb_	APB slave	Inbound
m_axi_desc_	AXI4 descriptor master	Outbound
m_axi_rd_	AXI4 read master	Outbound
m_axi_wr_	AXI4 write master	Outbound
monbus_	Monitor bus	Outbound

4.2 Diagram Conventions

4.2.1 Block Diagrams

- **Rectangles** - Functional blocks
- **Arrows** - Data flow direction
- **Dashed Lines** - Configuration/control paths
- **Double Lines** - High-bandwidth data paths

4.2.2 Timing Diagrams

- **High** - Logic 1
- **Low** - Logic 0
- **X** - Don't care / undefined
- **Hatched** - Valid data window

4.3 Numeric Conventions

4.3.1 Units

Unit	Meaning
Beat	Single data transfer (one clock cycle of valid data)

Unit	Meaning
Burst	Sequence of consecutive beats
Transaction	Complete AXI transaction (address + all data beats)

4.3.2 Address Notation

- All addresses are specified in bytes unless otherwise noted
- Address alignment requirements are specified relative to data width
- Example: 512-bit data width requires 64-byte (512/8) alignment

4.3.3 Size Notation

Notation	Value
KB	1,024 bytes
MB	1,048,576 bytes
Kbit	1,024 bits

4.4 Requirement Notation

Requirements and constraints are identified using the following format:

Prefix	Category
[REQ-x]	Functional requirement
[PERF-x]	Performance requirement
[INTEG-x]	Integration requirement
[CONST-x]	Design constraint

4.5 Reference Format

Cross-references to other documents use the format:

- **[MAS-x.y]** - Reference to MAS section x.y
- **[REF-n]** - Reference to document in reference table

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5 Definitions and Acronyms

5.1 Acronyms

Acronym	Definition
APB	Advanced Peripheral Bus
AXI	Advanced eXtensible Interface
AMBA	Advanced Microcontroller Bus Architecture
BFM	Bus Functional Model
CDC	Clock Domain Crossing
DMA	Direct Memory Access
FIFO	First-In First-Out buffer
FSM	Finite State Machine
HAS	Hardware Architecture Specification
MAS	Micro-Architecture Specification
MonBus	Monitor Bus (internal debug/trace interface)
PRD	Product Requirements Document
RTL	Register Transfer Level
SG	Scatter-Gather
SRAM	Static Random Access Memory
STREAM	Scatter-gather Transfer Rapid Engine for AXI Memory

5.2 Definitions

5.2.1 Architectural Terms

Term	Definition
Beat	A single data transfer on an AXI bus, corresponding to one clock cycle where valid data is transferred. For a 512-bit data bus, one beat transfers 64 bytes.
Burst	A sequence of consecutive beats forming a single AXI transaction. STREAM supports burst lengths up to 256 beats (AXI4 maximum).
Channel	An independent DMA transfer context. STREAM supports 8 channels, each capable of processing its own descriptor chain.
Descriptor	A 256-bit data structure in memory that defines a single DMA transfer operation, including source address, destination address, length, and chaining information.
Descriptor Chain	A linked list of descriptors where each descriptor points to the next, enabling complex scatter-gather transfer patterns.
Scatter-Gather	A DMA technique where data is transferred to/from non-contiguous memory locations using a list of descriptors.

5.2.2 Interface Terms

Term	Definition
APB Slave	STREAM's configuration interface, receiving programming commands from the system processor.
AXI Master	STREAM's memory interfaces for descriptor fetches and data transfers. Three masters: descriptor fetch, data read, data write.
MonBus	Monitor Bus - a 64-bit streaming interface for debug and performance monitoring packets.
Backpressure	Flow control mechanism where a receiver signals

Term	Definition
	inability to accept data, causing the sender to pause.

5.2.3 Transfer Terms

Term	Definition
Aligned Address	An address that is a multiple of the data width in bytes. For 512-bit (64-byte) data width, aligned addresses are multiples of 64.
Kick-off	Initiating a DMA transfer by writing the first descriptor address to a channel's control register.
Completion	The successful end of a descriptor chain, optionally generating an interrupt.
Chaining	Following the <code>next_descriptor_ptr</code> to fetch and process subsequent descriptors.

5.3 Units and Measurements

Term	Definition
Clock Cycle	One period of the primary system clock (<code>aclk</code>).
Latency	Time from initiating an operation to its completion, typically measured in clock cycles.
Throughput	Data transfer rate, typically measured in bytes per second or beats per cycle.
Utilization	Percentage of available bandwidth actually used for data transfer.

5.4 Document-Specific Terms

Term	Definition
External Interface	A signal or bus that crosses the STREAM subsystem boundary.
Internal Interface	A signal or bus between blocks within STREAM, not visible externally.
Configuration Register	An APB-accessible register for controlling STREAM

Term	Definition
Status Register	behavior. An APB-readable register reporting STREAM operational state.

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6 Use Cases

6.1 Primary Use Cases

6.1.1 Memory-to-Memory Data Movement

Description: Transfer data between two memory regions without CPU involvement.

Operation: 1. Software prepares descriptor(s) in memory with source/destination addresses 2. Software writes first descriptor address to channel control register 3. STREAM autonomously fetches descriptor and executes transfer 4. Optional interrupt notifies software of completion

Key Benefits: - CPU offloading for bulk data transfers - Deterministic transfer timing - Support for large transfer sizes via descriptor chaining

6.1.2 Scatter-Gather Operations

Description: Collect data from multiple non-contiguous source regions into a contiguous destination, or distribute contiguous source data to multiple destinations.

Gather Example:

```
Descriptor 0: src=0x1000, dst=0x8000, len=64 beats
Descriptor 1: src=0x3000, dst=0x8100, len=32 beats (chained)
Descriptor 2: src=0x5000, dst=0x8180, len=16 beats (chained, last)
```

Scatter Example:

Descriptor 0: src=0x8000, dst=0x1000, len=64 beats
Descriptor 1: src=0x8100, dst=0x3000, len=32 beats (chained)
Descriptor 2: src=0x8180, dst=0x5000, len=16 beats (chained, last)

6.1.3 Multi-Channel Concurrent Transfers

Description: Execute multiple independent transfer operations simultaneously.

Operation: - Up to 8 channels operate concurrently - Each channel processes its own descriptor chain - Shared resources (SRAM, AXI masters) arbitrated by priority

Use Cases: - Parallel DMA for multi-stream applications - Priority-based bandwidth allocation - Independent channel error isolation

6.2 Secondary Use Cases

6.2.1 Buffer Management

Description: Move data between system memory and local buffers for processing pipelines.

Pattern:

System Memory → STREAM → Processing Buffer → STREAM → System Memory

6.2.2 Memory Initialization

Description: Fill memory regions with pattern data using chained descriptors.

Note: Requires source buffer containing pattern data.

6.3 Target Applications

Application	Channel Usage	Typical Transfer Size
Video Frame Copy	1-2 channels	1-8 MB per frame
Audio Buffer Management	1 channel	4-64 KB per buffer
Data Collection	4-8 channels	Variable
Memory Test Patterns	1-4 channels	System-dependent

7 Key Features

7.1 Feature Summary

Feature	Specification
Channels	8 independent channels
Descriptor Size	256 bits (32 bytes)
Data Width	Parameterizable (default 512 bits)
Address Width	64 bits
Maximum Burst	256 beats (AXI4)
Chaining	Unlimited descriptor chain depth
Arbitration	Priority-based with round-robin

7.2 Descriptor-Based Operation

7.2.1 Single-Write Kick-off

A single APB write to a channel's control register initiates an entire transfer sequence:

1. Write descriptor address to CHn_CTRL register
2. STREAM automatically fetches and processes descriptor
3. Chained descriptors processed without software intervention
4. Completion indicated via status register and optional interrupt

7.2.2 Autonomous Chaining

STREAM follows next_descriptor_ptr automatically:

- **Non-zero pointer:** Fetch next descriptor, continue processing

- **Zero pointer:** Chain complete, transition to idle
 - **Last flag:** Explicit chain termination regardless of pointer
-

7.3 Multi-Channel Architecture

7.3.1 Channel Independence

Each channel maintains:

- Independent FSM state
- Separate descriptor chain pointer
- Individual error status
- Private completion interrupt

7.3.2 Resource Sharing

All channels share:

- Descriptor fetch AXI master
- Data read AXI master
- Data write AXI master
- Internal SRAM buffer
- MonBus reporter

7.3.3 Priority-Based Arbitration

- 8-bit priority field in each descriptor
 - Higher priority descriptors serviced first
 - Round-robin within same priority level
 - Anti-starvation timeout mechanism
-

7.4 Data Path Features

7.4.1 Parameterizable Data Width

Compile-time configurable data width:

Parameter	Description	Typical Values
DATA_WIDTH	AXI data bus width	128, 256, 512 bits

Parameter	Description	Typical Values
SRAM_DEPTH	Internal buffer depth	1024, 2048, 4096 entries

7.4.2 Aligned Address Requirement

[CONST-1] All source and destination addresses must be aligned to data width.

Data Width	Alignment Requirement
128 bits	16-byte aligned
256 bits	32-byte aligned
512 bits	64-byte aligned

This simplification eliminates complex alignment fixup logic for tutorial clarity.

7.5 Monitoring and Debug

7.5.1 MonBus Integration

64-bit monitor packets for:

- Transfer initiation events
- Descriptor fetch events
- Transfer completion events
- Error condition reporting
- Performance metrics

7.5.2 Status Visibility

APB-readable status registers provide:

- Channel state (idle, active, error)
- Descriptor count per channel
- Error codes and flags
- Performance counters (optional)

7.6 Error Handling

7.6.1 Error Detection

Error Type	Detection	Response
AXI read error	RRESP != OKAY	Stop chain, set error flag
AXI write error	BRESP != OKAY	Stop chain, set error flag
Invalid descriptor	Validation failure	Stop chain, set error flag
Address out of range	Limit check	Stop chain, set error flag

7.6.2 Error Isolation

Errors in one channel do not affect other channels. Each channel has independent error status and recovery.

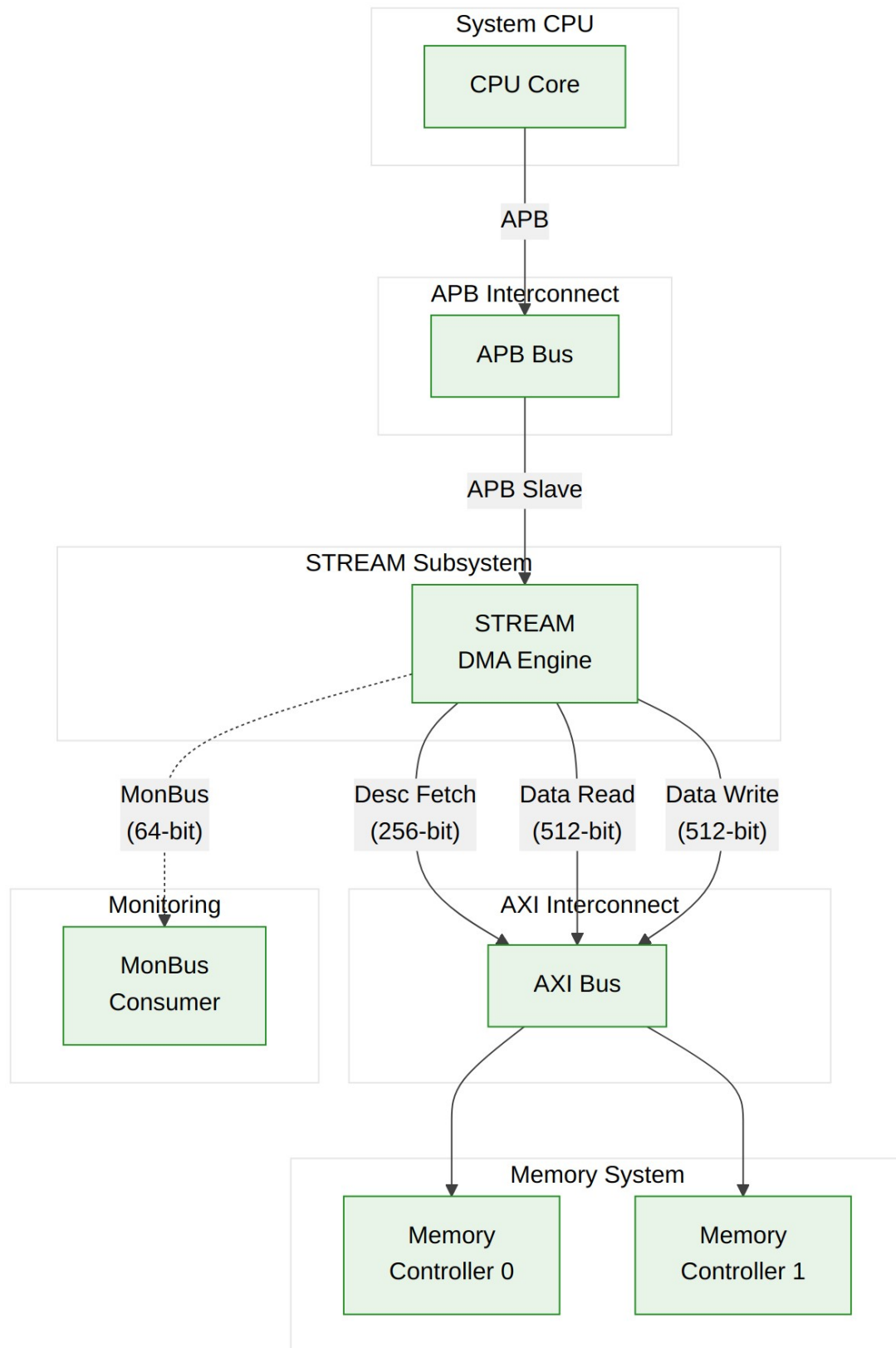
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8 System Context

8.1 System Integration Overview

STREAM integrates into a typical SoC as a peripheral subsystem with the following connections:



System Context Diagram

Source: [01_system_context.mmd](#)

8.2 External Connections

8.2.1 Configuration Path

Connection	Protocol	Description
APB Slave	APB3/APB4	Configuration and control interface

Functions: - Channel kick-off (write descriptor address) - Status monitoring - Interrupt control - Error handling

8.2.2 Data Paths

Connection	Protocol	Width	Description
Descriptor Master	AXI4	256-bit	Fetch descriptors from memory
Read Data Master	AXI4	Param	Read source data from memory
Write Data Master	AXI4	Param	Write destination data to memory

8.2.3 Monitoring Path

Connection	Protocol	Width	Description
MonBus Master	MonBus	64-bit	Debug/trace packet output

8.3 Clock and Reset

8.3.1 Clock Domain

STREAM operates in a single clock domain:

Signal	Description
aclk	Primary system clock, all interfaces synchronous
aresetn	Active-low asynchronous reset

[INTEG-1] All external interfaces must be synchronous to aclk.

8.3.2 Reset Requirements

- Asynchronous assertion, synchronous de-assertion
 - Minimum reset pulse width: 2 clock cycles
 - All state machines return to idle on reset
 - SRAM contents undefined after reset
-

8.4 Memory Map Requirements

8.4.1 Descriptor Memory

Descriptors must be placed in memory accessible via the descriptor fetch AXI master:

- **Alignment:** 32-byte aligned (256-bit descriptor size)
- **Accessibility:** Read access required
- **Coherency:** Software must ensure descriptors are coherent before kick-off

8.4.2 Data Memory

Source and destination regions must be accessible via the appropriate AXI master:

- **Source:** Read access via read data master
 - **Destination:** Write access via write data master
 - **Alignment:** Per data width (see Key Features)
-

8.5 Interrupt Integration

8.5.1 Interrupt Signals

Signal	Description
irq[7:0]	Per-channel completion/error interrupt
irq_combined	OR of all channel interrupts

8.5.2 Interrupt Conditions

- Descriptor with interrupt flag completes successfully
 - Channel encounters error condition
 - Software-triggered interrupt (via control register)
-

8.6 Power Considerations

8.6.1 Clock Gating

STREAM supports clock gating when all channels are idle:

- Monitor `all_channels_idle` status bit
- External clock gate can disable `aclk` when idle
- Re-enable clock before writing kick-off register

8.6.2 Power States

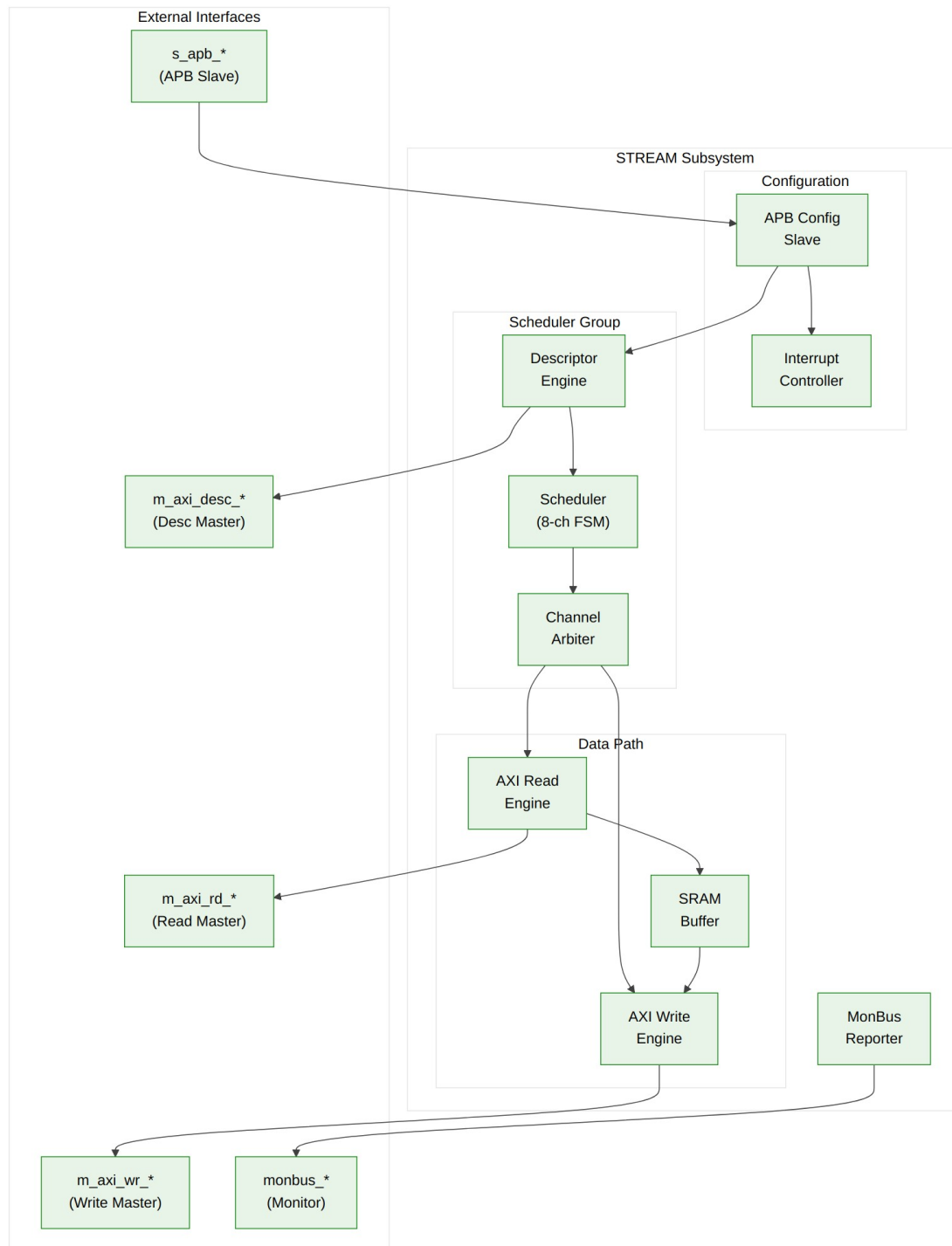
State	Channels	Clock	Power
Active	≥ 1 busy	Running	Full
Idle	All idle	Running	Reduced (no switching)
Gated	All idle	Stopped	Minimal (leakage only)

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9 Block Diagram

9.1 Top-Level Architecture



STREAM Block Diagram

Source: [02_block_diagram.mmd](#)

9.2 Functional Block Summary

Block	Function	Key Interfaces
APB Config Slave	Channel control registers, status	APB slave, kick-off signals
Descriptor Engine	Fetch and parse descriptors	AXI read, descriptor output
Scheduler	Coordinate transfer phases	Control signals to all blocks
Channel Arbiter	Multi-channel priority arbitration	Grant signals
AXI Read Engine	Read source data to SRAM	AXI read master
SRAM Buffer	Temporary data storage	Read/write ports
AXI Write Engine	Write SRAM data to destination	AXI write master
MonBus Reporter	Generate monitoring packets	MonBus output

9.3 Block Interactions

9.3.1 Control Flow

APB Write -> APB Config -> Descriptor Engine -> Scheduler -> Data Path, with AXI Desc Master fetching descriptors.

9.3.2 Data Flow

Memory -> AXI Read Engine -> SRAM Buffer -> AXI Write Engine -> Memory

9.3.3 Monitoring Flow

All Blocks -> Event Signals -> MonBus Reporter -> MonBus Output

9.4 Parameterization

Parameter	Description	Default	Range
NUM_CHANNELS	Number of DMA channels	8	1-8
DATA_WIDTH	AXI data bus width	512	128, 256, 512
ADDR_WIDTH	Address width	64	32, 64
SRAM_DEPTH	Buffer depth in entries	4096	256-8192
DESC_FIFO_DEPTH	Descriptor FIFO depth	4	2-8

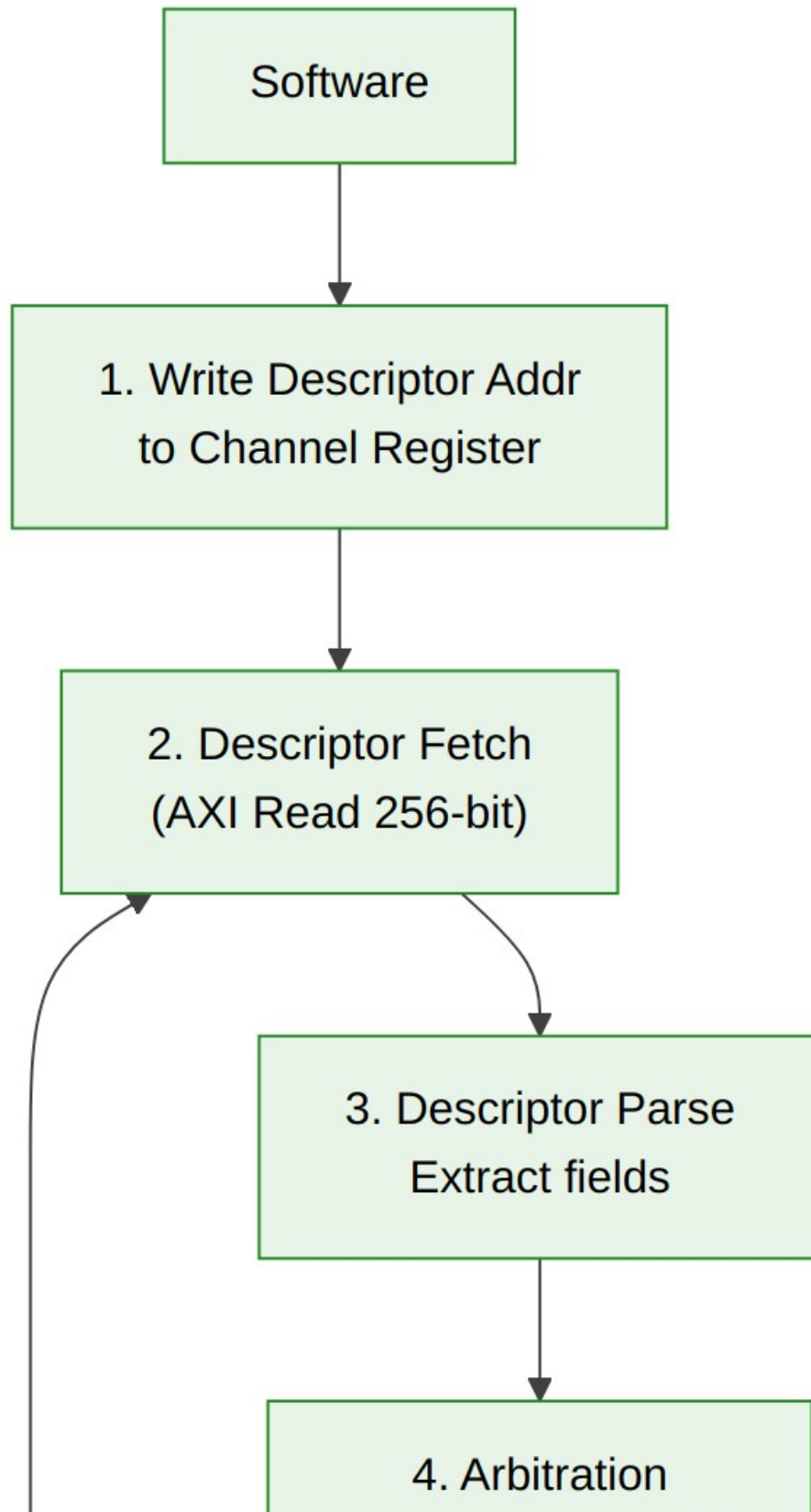
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10 Data Flow

10.1 Transfer Sequence Overview

A complete STREAM transfer follows this sequence:



10.2 Phase Details

10.2.1 Phase 1: Kick-off

Software writes the first descriptor address to the channel control register:

- APB write triggers channel state transition from IDLE to ACTIVE
- Descriptor address stored in channel's descriptor pointer register
- Kick-off signal sent to Descriptor Engine

10.2.2 Phase 2: Descriptor Fetch

Descriptor Engine issues AXI read transaction:

- Address: From kick-off or next_descriptor_ptr
- Size: 256 bits (single beat or 2-beat depending on bus width)
- Response: Descriptor data captured on valid read data

10.2.3 Phase 3: Descriptor Parse

Descriptor fields extracted and validated:

Field	Validation
valid	Must be 1
src_addr	Alignment check, range check
dst_addr	Alignment check, range check
length	Non-zero, within limits

10.2.4 Phase 4: Arbitration

Channel requests access to shared data path:

- Multiple channels may be active simultaneously
- Arbiter grants based on priority and fairness
- Granted channel proceeds to read phase

10.2.5 Phase 5: Read Phase

AXI Read Engine fetches source data:

- Issues AXI read bursts to source address
- Data written to SRAM buffer
- Continues until length beats transferred
- Read/write to SRAM can overlap (ping-pong or streaming)

10.2.6 Phase 6: Write Phase

AXI Write Engine sends data to destination:

- Reads data from SRAM buffer
- Issues AXI write bursts to destination address
- Awaits write responses
- Continues until all data written

10.2.7 Phase 7: Chain Check

Scheduler determines next action:

- If `last == 1` or `next_descriptor_ptr == 0`: Chain complete
 - Otherwise: Fetch next descriptor (return to Phase 2)
-

10.3 Concurrent Operation

10.3.1 Multi-Channel Concurrency

Multiple channels can be at different phases simultaneously. For example: - Ch0: Kick -> Fetch -> Read -> Write -> Complete - Ch1: (offset) Kick -> Fetch -> Arb Wait -> Read -> Write - Ch2: (offset) Kick -> Fetch -> Arb Wait -> Read

10.3.2 Pipeline Overlap

Within a single channel, read and write phases can overlap. As read beats 0-63 complete, write beats 0-63 can begin while read continues with beats 64-127.

The SRAM buffer enables this overlap by storing in-flight data.

10.4 Data Path Bandwidth

10.4.1 Theoretical Maximum

With 512-bit data width at 200 MHz:

- Per-direction: $512 \text{ bits} \times 200 \text{ MHz} = 102.4 \text{ Gbps} = 12.8 \text{ GB/s}$
- Bidirectional: 25.6 GB/s (simultaneous read + write)

10.4.2 Practical Considerations

Actual throughput affected by:

- AXI arbitration latency
 - Memory access latency
 - SRAM buffer depth (limits outstanding data)
 - Channel arbitration overhead
-

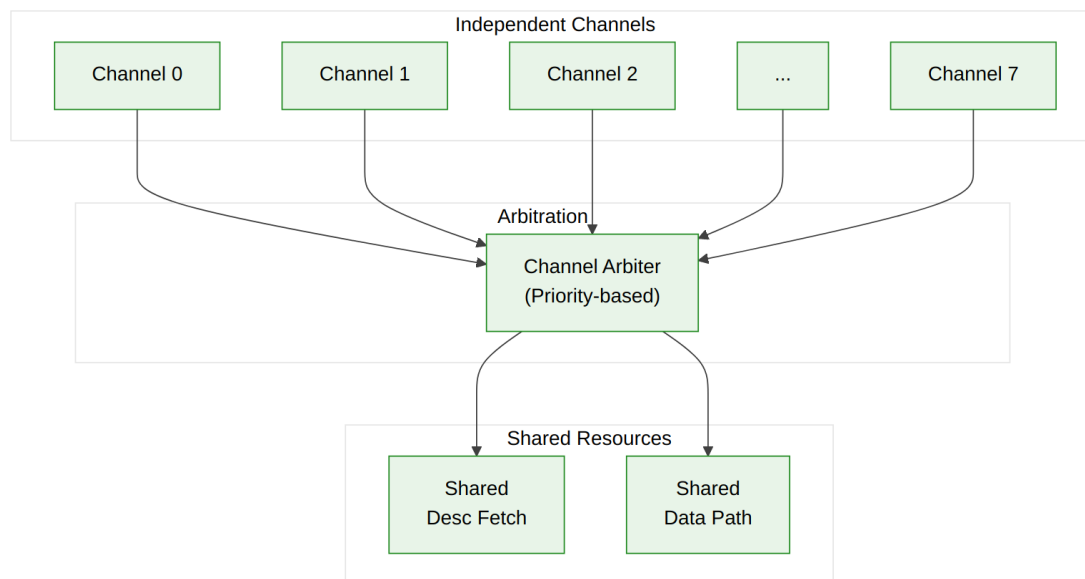
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11 Multi-Channel Architecture

11.1 Channel Organization

STREAM supports 8 independent DMA channels, each capable of processing its own descriptor chain:



Channel Architecture

Source: [04_channel_architecture.mmd](#)

11.2 Per-Channel Resources

Each channel maintains independent state:

Resource	Description	Size
FSM State	Current operation phase	4 bits
Descriptor Pointer	Current/next descriptor address	64 bits
Transfer Counter	Beats remaining in current transfer	32 bits
Error Status	Error flags and codes	8 bits
Priority	Current descriptor priority	8 bits

11.3 Shared Resources

All channels share the following resources:

Resource	Arbitration	Capacity
Descriptor AXI Master	Round-robin	1 outstanding transaction
Data Read AXI Master	Priority-based	Configurable outstanding
Data Write AXI Master	Priority-based	Configurable outstanding
SRAM Buffer	Partitioned by channel	SRAM_DEPTH entries
MonBus Reporter	FIFO-based	64-entry FIFO

11.4 Arbitration Scheme

11.4.1 Priority-Based Arbitration

Descriptor priority field (8 bits) determines service order:

- **Priority 255:** Highest priority
- **Priority 0:** Lowest priority
- **Same Priority:** Round-robin among equal priority

11.4.2 Fairness Mechanism

To prevent starvation:

- Maximum service time per grant (configurable)
- Anti-starvation counter promotes low-priority after timeout
- Emergency preemption for high-priority arrivals (optional)

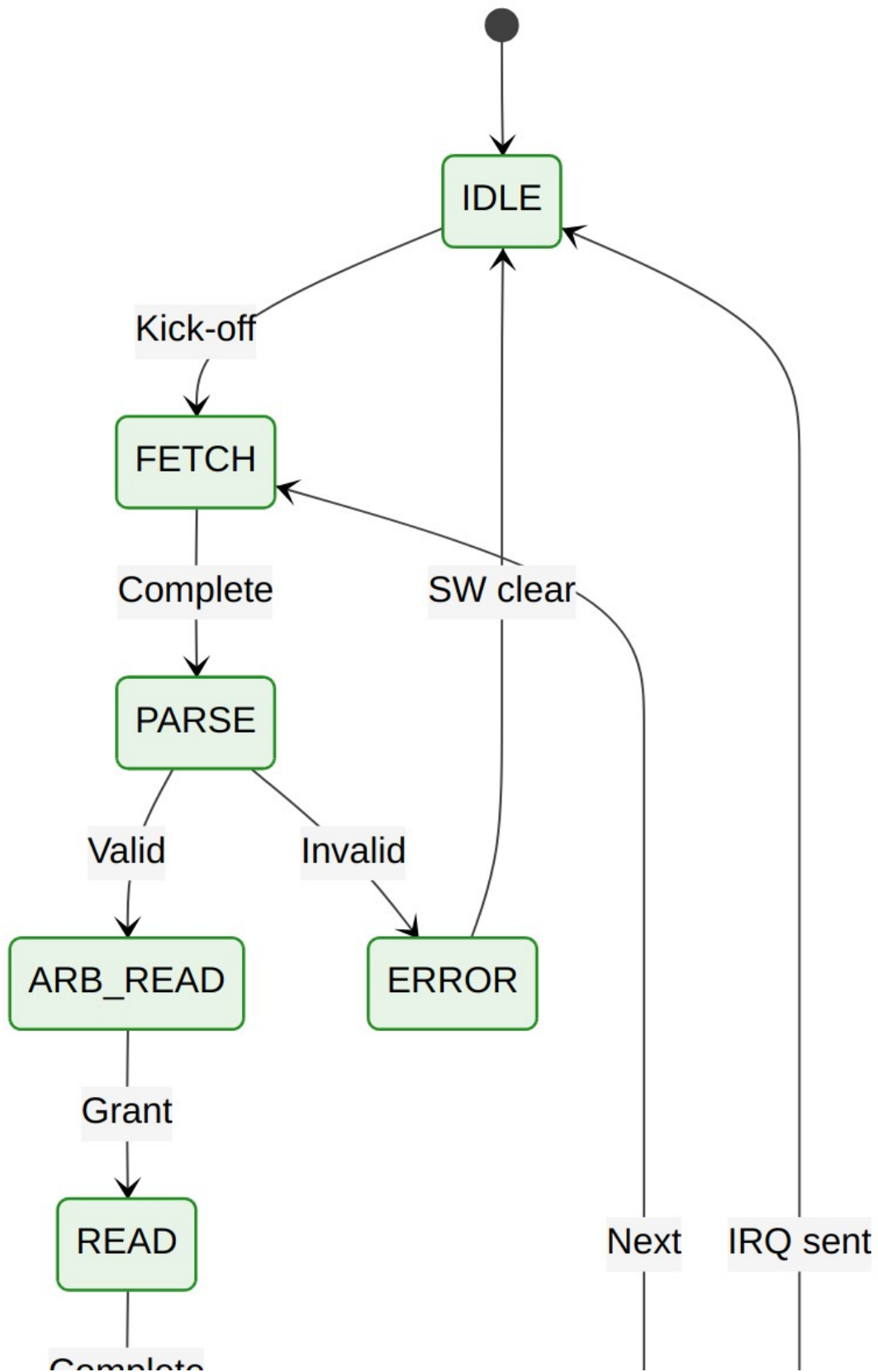
11.4.3 Arbitration Points

Arbitration occurs at:

1. **Descriptor Fetch:** When channel needs to fetch descriptor
2. **Read Phase Start:** When channel ready to read source data
3. **Write Phase Start:** When channel ready to write destination

11.5 Channel States

Each channel FSM traverses these states:



Channel FSM State Diagram

Source: [05_channel_fsm.mmd](#)

State	Description	Transitions
IDLE	No active transfer	Kick-off -> FETCH
FETCH	Fetching descriptor	Complete -> PARSE
PARSE	Validating descriptor	Valid -> ARB_READ, Invalid -> ERROR
ARB_READ	Awaiting read grant	Grant -> READ
READ	Reading source data	Complete -> ARB_WRITE
ARB_WRITE	Awaiting write grant	Grant -> WRITE
WRITE	Writing destination	Complete -> CHAIN
CHAIN	Checking for next	Next -> FETCH, Last -> COMPLETE
COMPLETE	Transfer done	IRQ sent -> IDLE
ERROR	Error condition	SW clear -> IDLE

11.6 SRAM Partitioning

The shared SRAM buffer is partitioned among active channels:

11.6.1 Static Partitioning (Default)

SRAM Buffer (4096 entries) is divided equally among 8 channels:

Partition	Entries
Channel 0	512
Channel 1	512
Channel 2	512
Channel 3	512
Channel 4	512
Channel 5	512
Channel 6	512
Channel 7	512

11.6.2 Dynamic Allocation (Optional)

When fewer than 8 channels are active, partitions can be expanded:

Active Channels	Entries Per Channel
1	4096
2	2048
4	1024
8	512

11.7 Error Isolation

Errors are isolated to the affected channel:

- Error in Channel N does not affect Channels 0..(N-1), (N+1)..7
- Shared resource errors logged but don't block other channels
- Software must clear error and re-kick to resume

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12 AXI Master Interfaces

12.1 Overview

STREAM includes three AXI4 master interfaces for memory access:

Interface	Purpose	Data Width	Address Width
m_axi_desc_*	Descriptor fetch	256 bits	64 bits
m_axi_rd_*	Source data read	Parameterizable	64 bits
m_axi_wr_*	Destination	Parameterizable	64 bits

Interface	Purpose	Data Width	Address Width
	write	le	

12.2 Descriptor Fetch Master

12.2.1 Signal Summary

Signal	Direction	Width	Description
m_axi_desc_ar_id	Output	ID_WIDTH	Read transaction ID
m_axi_desc_ar_addr	Output	64	Read address
m_axi_desc_ar_len	Output	8	Burst length (0 = 1 beat)
m_axi_desc_ar_size	Output	3	Beat size (5 = 32 bytes)
m_axi_desc_ar_burst	Output	2	Burst type (INCR)
m_axi_desc_ar_valid	Output	1	Address valid
m_axi_desc_ar_ready	Input	1	Address ready
m_axi_desc_rid	Input	ID_WIDTH	Read data ID
m_axi_desc_rdata	Input	256	Read data
m_axi_desc_resp	Input	2	Read response
m_axi_desc_rlast	Input	1	Last beat
m_axi_desc_rvalid	Input	1	Read data valid
m_axi_desc_rready	Output	1	Read data ready

12.2.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1 beat	Single 256-bit

Characteristic	Value	Notes
		descriptor
Burst Type	INCR	Fixed increment
Outstanding	1-2	Configurable
Ordering	In-order	Single ID used

12.2.3 Address Alignment

Descriptor addresses must be 32-byte aligned (256-bit boundary).

12.3 Data Read Master

12.3.1 Signal Summary

Signal	Direction	Width	Description
m_axi_rd_arid	Output	ID_WIDTH	Read transaction ID
m_axi_rd_aradr	Output	64	Read address
m_axi_rd_arlength	Output	8	Burst length
m_axi_rd_arsize	Output	3	Beat size
m_axi_rd_arburst	Output	2	Burst type (INCR)
m_axi_rd_arvalid	Output	1	Address valid
m_axi_rd_arready	Input	1	Address ready
m_axi_rd_rid	Input	ID_WIDTH	Read data ID
m_axi_rd_rdata	Input	DATA_WIDTH	Read data
m_axi_rd_rresp	Input	2	Read response
m_axi_rd_rlast	Input	1	Last beat
m_axi_rd_rvalid	Input	1	Read data valid

Signal	Direction	Width	Description
m_axi_rd_rready	Output	1	Read data ready

12.3.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1-256 beats	AXI4 maximum
Burst Type	INCR	Fixed increment
Outstanding	Configurable	4-16 typical
Ordering	In-order	Channel-based ID tagging

12.3.3 Burst Generation

Large transfers are broken into maximum-length bursts:

Transfer Size	Burst Count	Burst Lengths
256 beats	1	256
512 beats	2	256, 256
300 beats	2	256, 44

12.4 Data Write Master

12.4.1 Signal Summary

Signal	Direction	Width	Description
m_axi_wr_awid	Output	ID_WIDTH	Write transaction ID
m_axi_wr_awaddr	Output	64	Write address
m_axi_wr_awlength	Output	8	Burst length
m_axi_wr_awsize	Output	3	Beat size
m_axi_wr_awburst	Output	2	Burst type (INCR)
m_axi_wr_awvalid	Output	1	Address valid

Signal	Direction	Width	Description
m_axi_wr_awready	Input	1	Address ready
m_axi_wr_wdata	Output	DATA_WIDTH	Write data
m_axi_wr_wstrb	Output	DATA_WIDTH/8	Byte strobes
m_axi_wr_wlast	Output	1	Last beat
m_axi_wr_wvalid	Output	1	Write data valid
m_axi_wr_wready	Input	1	Write data ready
m_axi_wr_bid	Input	ID_WIDTH	Response ID
m_axi_wr_bresp	Input	2	Write response
m_axi_wr_bvalid	Input	1	Response valid
m_axi_wr_bready	Output	1	Response ready

12.4.2 Transaction Characteristics

Characteristic	Value	Notes
Burst Length	1-256 beats	Matches read bursts
Burst Type	INCR	Fixed increment
Outstanding	Configurable	4-16 typical
Write Strobes	All-ones	Aligned transfers only

12.5 Common AXI Attributes

12.5.1 Supported Features

Feature	Descriptor	Read	Write
INCR Burst	Yes	Yes	Yes
WRAP Burst	No	No	No

Feature	Descriptor	Read	Write
FIXED Burst	No	No	No
Exclusive	No	No	No
Locked	No	No	No

12.5.2 Error Handling

- SLVERR/DECERR on any channel stops the current transfer
- Error logged in channel status register
- Subsequent descriptors in chain not processed
- Software intervention required to clear and resume

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13 APB Configuration Interface

13.1 Overview

STREAM provides an APB slave interface for software configuration and control. The interface supports:

- Channel kick-off (write descriptor address to start transfer)
- Status monitoring
- Interrupt control
- Error handling

13.2 Signal Summary

Signal	Direction	Width	Description
s_apb_paddr	Input	13	Address bus (8 KB space; reaches monitor block at 0x1000)

Signal	Direction	Width	Description
s_apb_psel	Input	1	Peripheral select
s_apb_penable	Input	1	Enable phase
s_apb_pwrite	Input	1	Write enable
s_apb_pwdata	Input	32	Write data
s_apb_pstrb	Input	4	Byte strobes
s_apb_pready	Output	1	Ready signal
s_apb_prdata	Output	32	Read data
s_apb_pslverr	Output	1	Slave error

13.3 Register Map Overview

Offset	Register	Access	Description
0x000	CTRL	RW	Global control register
0x004	STATUS	RO	Global status register
0x008	IRQ_EN	RW	Interrupt enable mask
0x00C	IRQ_STATUS	RW1C	Interrupt status
0x010	ERR_STATUS	RO	Error status
0x014	ERR_ADDR	RO	Error address capture
0x020-0x03C	Reserved	-	Reserved
0x040	CH0_CTRL	RW	Channel 0 control
0x044	CH0_STATUS	RO	Channel 0 status
0x048	CH0_DESC_PTR	RO	Channel 0 current descriptor
0x04C	Reserved	-	Reserved
0x050	CH1_CTRL	RW	Channel 1 control
...	...	...	...
0x0B0	CH7_CTRL	RW	Channel 7 control
0x0B4	CH7_STATUS	RO	Channel 7 status
0x0B8	CH7_DESC_PTR	RO	Channel 7 current descriptor

Offset	Register	Access	Description
0x0BC	Reserved	-	Reserved
0x0C0	OBS_CTRL	RW	Channel observation mux select
0x0C4	OBS_FLAGS	RO	Observation flags for selected channel
0x0C8	OBS_DATA0	RO	Observation data 0 (low)
0x0CC	OBS_DATA1	RO	Observation data 1 (high)
0x1000+	Monitor / performance registers	RW/RO	AXI monitor + per-monitor performance registers (separate stream_mon_regs regfile at 0x1000, same APB slave)

Note: The AXI monitor and performance registers live in a separate `stream_mon_regs` regfile instantiated at offset **0x1000** on the same APB slave. The APB address decode is therefore **13 bits (8 KB)** so those registers are addressable. Detailed monitor register layout is in the MAS (`stream_mas/ch04_registers/register_map.md`).

13.4 Key Registers

13.4.1 Global Control (CTRL) - 0x000

Bits	Field	Access	Description
[0]	ENABLE	RW	Global enable
[1]	SOFT_RESET	RW	Soft reset (self-clearing)
[31:2]	Reserved	-	Reserved

13.4.2 Global Status (STATUS) - 0x004

Bits	Field	Access	Description
[0]	BUSY	RO	Any channel active
[7:1]	Reserved	-	Reserved
[15:8]	CH_ACTIVE	RO	Per-channel

Bits	Field	Access	Description
[23:16]	CH_ERROR	RO	active mask Per-channel error mask
[31:24]	Reserved	-	Reserved

13.4.3 Channel Control (CHn_CTRL) - 0x040 + n*0x10

Bits	Field	Access	Description
[31:0]	DESC_ADDR	RW	Descriptor address (kick-off on write)

Writing to CHn_CTRL initiates a transfer: 1. Descriptor address stored 2. Channel transitions from IDLE to FETCH 3. PREADY de-asserted until channel accepts kick-off

13.4.4 Channel Status (CHn_STATUS) - 0x044 + n*0x10

Bits	Field	Access	Description
[3:0]	STATE	RO	Channel FSM state
[4]	IDLE	RO	Channel is idle
[5]	BUSY	RO	Channel is processing
[6]	ERROR	RO	Error condition
[7]	COMPLETE	RO	Last transfer complete
[15:8]	DESC_COUNT	RO	Descriptors processed
[23:16]	ERR_CODE	RO	Error code if ERROR set
[31:24]	Reserved	-	Reserved

13.4.5 Channel Observation Control (OBS_CTRL) - 0x0C0

Bits	Field	Access	Description
[2:0]	CHANNEL_SEL	RW	Channel index (0-7) for observation read

Bits	Field	Access	Description
[31:3]	Reserved	-	Reserved

Software writes the channel index to this register, then reads OBS_FLAGS and OBS_DATA0/DATA1 to observe that channel's internal state.

13.4.6 Channel Observation Flags (OBS_FLAGS) - 0x0C4

Bits	Field	Access	Description
[0]	SCH_ERROR_STICKY	RO	Scheduler error sticky (descriptor invalid, timeout, etc.)
[1]	SCH_TIMEOUT	RO	Scheduler timeout state (waiting on AXI master)
[7:2]	Reserved	-	Reserved
[31:8]	Reserved	-	Reserved

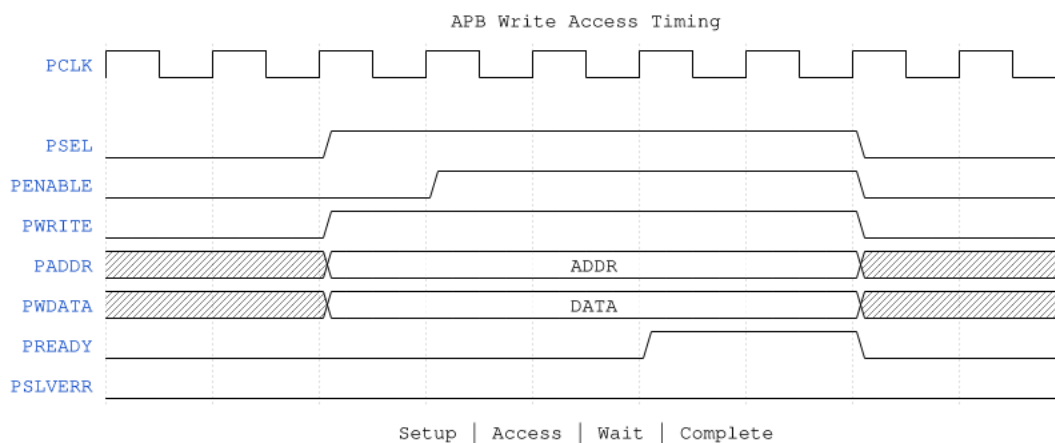
These flags are driven by the scheduler's internal error and timeout monitoring. See `stream_mas/ch02_blocks/04_scheduler.md` for detailed definitions.

13.4.7 Channel Observation Data (OBS_DATA0, OBS_DATA1) - 0x0C8, 0x0CC

Per-channel data such as descriptor state, current beat count, or other diagnostic fields. Exact contents vary by channel state; see the register map audit document for current field assignments.

13.5 Access Timing

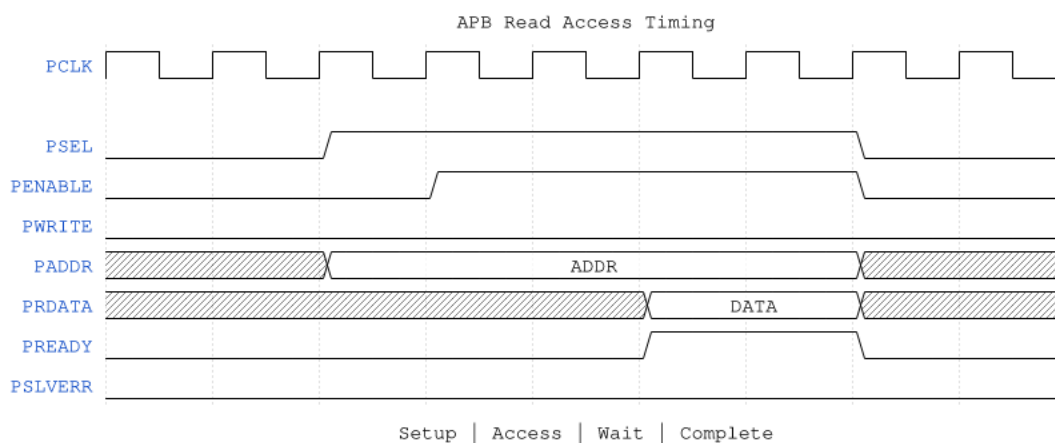
13.5.1 Write Access



APB Write Access Timing

Source: [apb_write_access.json](#)

13.5.2 Read Access



APB Read Access Timing

Source: [apb_read_access.json](#)

13.6 Kick-off Blocking

When software writes to CHn_CTRL:

- If channel is IDLE: Immediate accept, PREADY asserted
- If channel is BUSY: PREADY held low until channel completes
- Prevents descriptor pointer corruption during active transfer

13.7 Kick-Burst Fast-Path Interface (optional)

In addition to the APB CHn_CTRL kick, the top level exposes a **kick-burst** interface — a set of top-level ports (NOT APB registers) that let an integrator start any subset of channels **on a single clock cycle**, bypassing the per-channel APB write sequence. This matters when the APB path is reached over a slow transport (e.g. a UART-to-APB bridge on an FPGA), where issuing eight separate two-register kicks would serialize channel starts over milliseconds.

Kick-Burst Interface Signals

Signal	Direction	Width	Description
i_kick_burst_mask	Input	NUM_CHANNEL S	One-cycle pulse per channel; a set bit enqueues a kick for that channel
i_kick_burst_addr	Input	NUM_CHANNEL S x ADDR_WIDTH	Per-channel descriptor address, sampled when the matching mask bit pulses

Semantics. On any cycle where i_kick_burst_mask[ch] is high, the descriptor address on i_kick_burst_addr[ch] is latched into a per-channel pending kick and delivered to that channel's scheduler as soon as it can accept it. Asserting several mask bits on the same cycle enqueues all of those channels' kicks together, so N channels start back-to-back rather than serially. The pending kick clears once the scheduler accepts it.

The kick-burst path and the APB CHn_CTRL path feed the same per-channel kick input (OR-combined; the APB path wins the address if both fire on one cycle, which is not expected in practice). An integrator drives i_kick_burst_* from its own logic — typically a bank of per-channel address shadow registers plus a “go” register whose write pulses the mask. The NexysA7 characterization harness is the reference implementation (address shadow registers + a KICK_G0 bitmask CSR); see the MAS programming chapter for the software sequence.

Note: The kick-burst ports are unrelated to the APB address map above. In a build that does not use them, tie `i_kick_burst_mask` low.

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14 Monitoring Interface

14.1 Overview

STREAM provides a unified Monitor Bus (MonBus) interface for debug and performance monitoring. This interface streams event packets (128-bit) with paired 64-bit side-band timestamps, capturing key operational events from all internal blocks.

14.2 Signal Summary

Signal	Direction	Width	Description
monbus_pkt_valid	Output	1	Packet valid
monbus_pkt_ready	Input	1	Downstream ready
monbus_pkt_data	Output	128	Packet data (protocol-independent monitor packet)
monbus_ts_data	Output	64	Side-band timestamp (cycle count at packet capture)

14.3 Packet Format

The 128-bit MonBus packet and 64-bit timestamp are paired atomically and transmitted as a 192-bit group through AXI-Lite skid buffering. See the canonical packet specification at

docs/markdown/RTLamba/includes/monitor_package_spec.md for complete field definitions.

Bits	Field	Description
[63:56]	SUBSYSTEM_ID	STREAM subsystem identifier
[55:48]	EVENT_TYPE	Event category
[47:40]	EVENT_CODE	Specific event within category
[39:32]	CHANNEL_ID	Associated channel (0-7)
[31:0]	PAYLOAD	Event-specific data

14.4 Monitoring Architecture

14.4.1 Descriptor Path Monitoring (Both Ends)

STREAM monitors the descriptor fetch path at **both ends** for observability:

1. **Descriptor Fetch Master (Agent ID 8):** Monitors the AXI4 master that fetches descriptors from system memory
2. **Descriptor Consume Side (Agent ID 48 + channel):** Monitors when the scheduler consumes the descriptor for processing

This dual-end monitoring allows software to attribute descriptor path stalls to either fetch delay (master side) or consume delay (slave side), enabling faster root-cause analysis.

14.4.2 Global Monitor Enable Methodology

All internal monitors are gated by: - **Global SV parameter:** USE_MONITOR (default: enabled for analysis, disabled for area-optimized builds) - **Per-port SV parameters:** Individual monitor enable for each monitored port - **APB runtime configuration:** Masks for enabling/disabling packet types at run time

14.5 Event Types

STREAM uses the unified MonBus event categorization defined in docs/markdown/RTLamba/includes/monitor_package_spec.md: - **Error events (Type 0x0):** Protocol violations, timeouts, invalid descriptors - **Completion events (Type 0x1):** Successful

transfer completion, descriptor processed - **Performance events (Type 0x4)**: Latency metrics, throughput data - **Protocol-specific events**: AXI address-match, debug traces

All STREAM events use **Unit ID = 1** and channel-specific **Agent IDs** for the source block (descriptor engine, scheduler, AXI monitor, etc.).

Do not re-state the 128-bit packet layout here — refer to the canonical specification for field definitions and bit assignments.

14.6 Event Configuration

14.6.1 Configurable Events

Events can be enabled/disabled via APB registers:

Register	Bit	Event Category
MONBUS_CFG	[0]	Transfer events
MONBUS_CFG	[1]	Error events
MONBUS_CFG	[2]	Performance events
MONBUS_CFG	[7:4]	Channel mask

14.6.2 Event Priority

When multiple events occur simultaneously:

1. Error events (highest priority)
 2. Transfer completion events
 3. Transfer start events
 4. Performance events (lowest priority)
-

14.7 Backpressure Handling

14.7.1 Flow Control

- Events generated when monbus_pkt_ready is asserted
- If monbus_pkt_ready is low, events are queued in internal FIFO
- FIFO depth: 64 entries (configurable)
- Overflow behavior: Oldest events discarded, overflow counter incremented

14.7.2 Recommendations

- Connect MonBus to FIFO with sufficient depth
 - Monitor overflow counter in status register
 - Size downstream FIFO based on expected event rate
-

14.8 Typical MonBus Downstream

STREAM MonBus → MonBus FIFO → MonBus Arbiter → System Trace
(64 deep) (multi-source) Interface

14.9 Example Event Sequence

A single-descriptor transfer generates these events:

Time	Event	
----	-----	
T0	KICKOFF	- Software kicks off channel
T1	DESC_FETCH	- Descriptor fetch initiated
T10	DESC_VALID	- Descriptor parsed successfully
T12	ARB_WAIT	- Waiting for data path grant (if applicable)
T15	READ_START	- AXI read burst initiated
T75	READ_DONE	- All source data read
T80	WRITE_START	- AXI write burst initiated
T140	WRITE_DONE	- All data written, responses received
T141	COMPLETE	- Channel complete, back to IDLE

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15 Throughput Targets

15.1 Theoretical Maximum

15.1.1 Per-Interface Bandwidth

Interface	Data Width	Frequency	Bandwidth
Descriptor Fetch	256 bits	200 MHz	6.4 GB/s
Data Read	512 bits	200 MHz	12.8 GB/s
Data Write	512 bits	200 MHz	12.8 GB/s

15.1.2 Aggregate Bandwidth

- **Bidirectional:** 25.6 GB/s (simultaneous read + write)
- **Descriptor Overhead:** ~0.1% for large transfers

15.2 Practical Throughput

15.2.1 Single-Channel Performance

Scenario	Expected Efficiency	Effective Bandwidth
Large sequential (>1MB)	>95%	>12.0 GB/s
Medium sequential (64KB-1MB)	85-95%	10.8-12.0 GB/s
Small sequential (4KB-64KB)	70-85%	9.0-10.8 GB/s
Very small (<4KB)	40-70%	5.0-9.0 GB/s

15.2.2 Multi-Channel Performance

Active Channels	Per-Channel BW	Aggregate BW
1	12.8 GB/s	12.8 GB/s
2	6.4 GB/s	12.8 GB/s
4	3.2 GB/s	12.8 GB/s
8	1.6 GB/s	12.8 GB/s

Note: Aggregate limited by shared AXI masters. Individual channel throughput scales inversely with active channel count.

15.3 Throughput Limiting Factors

15.3.1 Memory System Factors

Factor	Impact	Mitigation
Memory latency	Reduces efficiency for small transfers	Use deeper outstanding
Memory bandwidth	Hard limit	Match STREAM to memory capability
Interconnect contention	Variable impact	Priority configuration

15.3.2 STREAM Internal Factors

Factor	Impact	Mitigation
Descriptor fetch overhead	Fixed per descriptor	Use longer transfers
Per-descriptor boundary bubble	~13 cyc/descriptor (chains)	Read-ahead prefetch (default on) — eliminates it
SRAM depth	Limits outstanding	Configure adequate depth
Arbitration overhead	Multi-channel penalty	Reduce active channels
Burst splitting	4KB boundary crossings	Align transfers

15.4 Performance Targets

15.4.1 Design Targets

Metric	Target	Condition
Peak throughput	12.8 GB/s	Single direction, ideal
Sustained throughput	>11.0 GB/s	Large transfers, single channel

Metric	Target	Condition
Multi-channel aggregate	>10.0 GB/s	8 channels active
Small transfer efficiency	>50%	4KB transfers

15.4.2 Verification Criteria

Scenario	Pass Criteria
1MB sequential read	>95% efficiency
1MB sequential write	>95% efficiency
64KB scatter (4x16KB)	>80% efficiency
4KB × 100 descriptors	>60% efficiency

15.5 Descriptor Chain Efficiency

15.5.1 Descriptor Overhead

Transfer Size	Descriptor Fetches	Overhead
64 KB	1	0.05%
256 KB	1	0.01%
1 MB	1	<0.01%
4 KB (100 chained)	100	0.5%

15.5.2 Chaining Efficiency

For chained descriptors, next descriptor fetch can overlap with current transfer:

```

Desc 0: [Fetch][-----Transfer 0-----]
Desc 1:           [Fetch][-----Transfer 1-----]
Desc 2:                           [Fetch][-----Transfer 2-----]
```

This pipelining minimizes chaining overhead for continuous transfers.

15.5.3 Read-Ahead Descriptor Prefetch (Cross-Descriptor Streaming)

Descriptor fetch overlaps the current transfer (above), but the datapath itself must also stream across the boundary. Without prefetch the read side stalls at each descriptor boundary while the scheduler FSM advances and the read pipeline (AR → R) refills — a per-descriptor **bubble** of ~13 cycles that dominates on short-descriptor chains. Read-ahead descriptor prefetch (runtime `SCHED_CONFIG.RD_PREFETCH_EN`, default **on**; see MAS Section 2.4) advances the read side to

descriptor N+1 while the write side still drains N, so the read stream never goes cold. The per-descriptor bubble drops to **zero** — chain throughput becomes independent of descriptor count, with only a one-time fill/drain of overhead.

Measured on silicon (Nexys A7, 100 MHz, one bitstream, prefetch on vs off):

Descriptor size	Prefetch OFF (lockstep)	Prefetch ON
64 beats (16-desc chain)	76.1% datapath util	95.3%
>= 4096 beats	~99%	99-100%

Applies to chained **legacy** descriptors. EXT (row/col-major) descriptors run lockstep, but a single row-major descriptor is internally bubble-free (its contiguous runs stream through the address-generator base FIFO — measured 99.7% util). Transpose access patterns (col-major on either side) are single-beat and latency-bound by the memory system, not by a scheduler bubble.

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16 Latency Characteristics

16.1 Latency Components

16.1.1 End-to-End Latency Breakdown

For a single-descriptor transfer (kick-off to completion):

Phase	Typical Cycles	Notes
APB write processing	2-3	APB transaction
Descriptor fetch	10-50	Memory latency dependent
Descriptor parse	2-4	Fixed pipeline
Arbitration wait	0-100+	Depends on contention

Phase	Typical Cycles	Notes
Read phase setup	2-3	AXI AR channel
Read data transfer	N + 10-50	N beats + memory latency
Write phase setup	2-3	AXI AW channel
Write data transfer	N + 5-20	N beats + response latency
Completion handling	2-4	Status update, IRQ

16.1.2 Total Latency Formula

$$\text{Latency} = T_{\text{apb}} + T_{\text{desc_fetch}} + T_{\text{parse}} + T_{\text{arb}} + T_{\text{read}} + T_{\text{write}} + T_{\text{complete}}$$

Where:

$$\begin{aligned}
 T_{\text{apb}} &= 3 \text{ cycles (fixed)} \\
 T_{\text{desc_fetch}} &= \text{Memory_latency} + 10 \text{ cycles} \\
 T_{\text{parse}} &= 3 \text{ cycles (fixed)} \\
 T_{\text{arb}} &= 0 \text{ to } N \text{ (contention dependent)} \\
 T_{\text{read}} &= \text{Memory_latency} + (\text{Transfer_length} / \text{Beats_per_burst}) \\
 &+ \text{Burst_overhead} + \text{Transfer_length} \\
 T_{\text{write}} &= \text{Memory_latency} + \text{Transfer_length} + \text{Response_latency} \\
 T_{\text{complete}} &= 3 \text{ cycles (fixed)}
 \end{aligned}$$

16.2 First-Byte Latency

Time from kick-off to first data beat arriving at destination:

Component	Cycles	Cumulative
APB processing	3	3
Descriptor fetch	30	33
Parse + setup	5	38
Read AR accepted	10	48
First read data	20	68
Write AW + first W	10	78
Total	~80	-

At 200 MHz: ~400 ns first-byte latency (typical)

16.3 Latency by Transfer Size

16.3.1 Small Transfers

Transfer Size	Total Latency	Efficiency
64 bytes (1 beat)	~100 cycles	1%
256 bytes (4 beats)	~110 cycles	4%
1 KB (16 beats)	~130 cycles	12%
4 KB (64 beats)	~180 cycles	35%

16.3.2 Large Transfers

Transfer Size	Total Latency	Efficiency
16 KB (256 beats)	~400 cycles	64%
64 KB (1024 beats)	~1200 cycles	85%
256 KB (4096 beats)	~4400 cycles	93%
1 MB (16384 beats)	~17000 cycles	96%

16.4 Latency Variability

16.4.1 Sources of Variability

Source	Impact	Range
Memory system load	High	10-100+ cycles
AXI interconnect	Medium	5-50 cycles
Channel arbitration	Medium	0-100+ cycles
Descriptor chaining	Low	2-5 cycles

16.4.2 Worst-Case Latency

Conditions for worst-case latency:

1. All 8 channels active (maximum arbitration delay)
2. Memory system fully loaded (maximum memory latency)
3. Small transfer size (maximum overhead ratio)
4. Low priority descriptor (maximum wait time)

Under these conditions, first-byte latency could exceed 1000 cycles.

16.5 Latency Optimization

16.5.1 Software Recommendations

Recommendation	Benefit
Use larger transfers	Better efficiency
Minimize active channels	Reduce arbitration
Use higher priority for latency-sensitive	Earlier service
Pre-stage descriptors in cache	Faster fetch

16.5.2 Hardware Configuration

Setting	Effect
Increase outstanding limit	Hide memory latency
Enable descriptor prefetch	Reduce chaining overhead
Configure larger SRAM	Support longer bursts

16.6 Interrupt Latency

Time from transfer completion to interrupt assertion:

Phase	Cycles
Last write response	0
Completion FSM	2
Status register update	1
Interrupt assertion	1
Total	4 cycles

At 200 MHz: 20 ns from completion to interrupt.

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17 Resource Estimates

17.1 FPGA Resource Summary

Estimates for typical configuration (8 channels, 512-bit data width, 4096-entry SRAM):

Resource	Estimate	Notes
LUTs	15,000 - 25,000	Logic and control
Registers	20,000 - 30,000	Pipeline and state
Block RAM	40 - 60	SRAM buffer
DSP	0	No DSP usage

17.2 Resource Breakdown by Block

17.2.1 APB Configuration Slave

Resource	Estimate
LUTs	500 - 800
Registers	400 - 600
BRAM	0

17.2.2 Descriptor Engine

Resource	Estimate
LUTs	1,500 - 2,500
Registers	1,000 - 1,500
BRAM	1 (descriptor FIFO)

17.2.3 Scheduler (8 channels)

Resource	Estimate
LUTs	3,000 - 5,000
Registers	4,000 - 6,000

Resource	Estimate
BRAM	0

17.2.4 Channel Arbiter

Resource	Estimate
LUTs	500 - 1,000
Registers	200 - 400
BRAM	0

17.2.5 AXI Read Engine

Resource	Estimate
LUTs	2,000 - 3,500
Registers	3,000 - 4,500
BRAM	0

17.2.6 SRAM Buffer

Resource	Estimate
LUTs	100 - 200
Registers	100 - 200
BRAM	32 - 64 (depth/width dependent)

17.2.7 AXI Write Engine

Resource	Estimate
LUTs	2,000 - 3,500
Registers	3,000 - 4,500
BRAM	0

17.2.8 MonBus Reporter

Resource	Estimate
LUTs	800 - 1,200
Registers	600 - 900
BRAM	1 (event FIFO)

17.3 SRAM Sizing

SRAM buffer size depends on configuration:

Configuration	BRAM (18Kb)	BRAM (36Kb)
512b x 1024	16	8
512b x 2048	32	16
512b x 4096	64	32
256b x 4096	32	16
128b x 4096	16	8

17.4 Scaling with Parameters

17.4.1 Channel Count Impact

Channels	LUT Delta	Register Delta
1	Base	Base
2	+500	+800
4	+1,500	+2,400
8	+3,500	+5,600

17.4.2 Data Width Impact

Data Width	LUT Delta	Register Delta	BRAM Delta
128 bits	-30%	-40%	-75%
256 bits	-15%	-20%	-50%
512 bits	Base	Base	Base

17.5 ASIC Estimates

For 28nm technology node (typical):

Metric	Estimate
Gate Count	200K - 350K gates
Area	0.2 - 0.4 mm ²
SRAM Area	0.1 - 0.2 mm ² (4096x512)

Metric	Estimate
Total Area	0.3 - 0.6 mm ²

17.5.1 Power Estimates

Condition	Power
Idle (clock gated)	<1 mW
Idle (clock running)	10 - 20 mW
Single channel active	50 - 100 mW
All channels active	150 - 300 mW

17.6 Timing Estimates

17.6.1 Target Frequencies

Technology	Target Fmax
Xilinx Ultrascale+	250 MHz
Intel Agilex	250 MHz
28nm ASIC	400 MHz
16nm ASIC	600 MHz

17.6.2 Critical Paths

Path	Concern Level	Mitigation
AXI data path	Low	Fully pipelined
SRAM read/write	Low	Registered outputs
Arbiter decision	Medium	Priority encoder depth
Descriptor parse	Low	Simple combinational

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18 System Requirements

18.1 Interface Requirements

18.1.1 APB Interface

Requirement	Specification
Protocol	APB3 or APB4
Data Width	32 bits
Address Width	12 bits minimum
Timing	Synchronous to ac1k

18.1.2 AXI Interfaces

Requirement	Descriptor	Read Data	Write Data
Protocol	AXI4	AXI4	AXI4
Data Width	256 bits	Parameterizable	Parameterizable
Address Width	64 bits	64 bits	64 bits
ID Width	4 bits min	4 bits min	4 bits min
Outstanding	2	Configurable	Configurable

18.1.3 MonBus Interface

Requirement	Specification
Data Width	64 bits
Protocol	Valid/Ready streaming
Backpressure	Required (must handle ready deassertion)

18.2 Memory Requirements

18.2.1 Descriptor Memory

Requirement	Specification
Alignment	32-byte aligned
Access	Read access via descriptor AXI master
Coherency	Software responsibility
Latency	<100 cycles recommended

18.2.2 Data Memory

Requirement	Specification
Alignment	Per data width (64-byte for 512-bit)
Access	Read via read master, Write via write master
Bandwidth	Sufficient for target throughput

18.3 System Bandwidth Allocation

18.3.1 AXI Interconnect Requirements

To achieve maximum STREAM throughput:

AXI Master	Required Bandwidth	Priority
Descriptor Fetch	0.1 GB/s	Medium
Data Read	12.8 GB/s	High
Data Write	12.8 GB/s	High

18.3.2 Interconnect Recommendations

- Configure STREAM AXI masters with high priority
 - Ensure sufficient outstanding transaction support
 - Consider dedicated paths for high-bandwidth DMA
-

18.4 Interrupt Requirements

18.4.1 Interrupt Controller

Requirement	Specification
Interrupt Lines	8 (per-channel) or 1 (combined)
Type	Level-sensitive, active-high
Latency	<10 cycles from assertion to controller

18.4.2 Interrupt Handling

- Software must read IRQ_STATUS to identify source
 - Write-1-to-clear to acknowledge interrupts
 - Channel remains blocked until error cleared (if applicable)
-

18.5 Address Space Requirements

18.5.1 STREAM Register Space

Requirement	Specification
Base Address	System-defined
Size	4 KB (0x1000)
Alignment	4 KB aligned

18.5.2 Memory Map Constraints

- Descriptor addresses must be accessible via descriptor AXI master
 - Source addresses must be accessible via read AXI master
 - Destination addresses must be accessible via write AXI master
 - All addresses must respect configured address range limits
-

18.6 Software Requirements

18.6.1 Driver Requirements

Requirement	Description
Descriptor Preparation	Build descriptors in coherent

Requirement	Description
	memory
Cache Management	Flush descriptor cache before kick-off
Interrupt Handling	Service interrupts promptly
Error Recovery	Clear error status, optionally retry

18.6.2 Programming Sequence

1. Verify STREAM is enabled (`CTRL.ENABLE = 1`)
2. Check target channel is idle (`CHn_STATUS.IDLE = 1`)
3. Prepare descriptor chain in memory
4. Flush cache (if applicable)
5. Write first descriptor address to `CHn_CTRL`
6. Wait for interrupt or poll `CHn_STATUS`

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19 Clocking and Reset

19.1 Clock Requirements

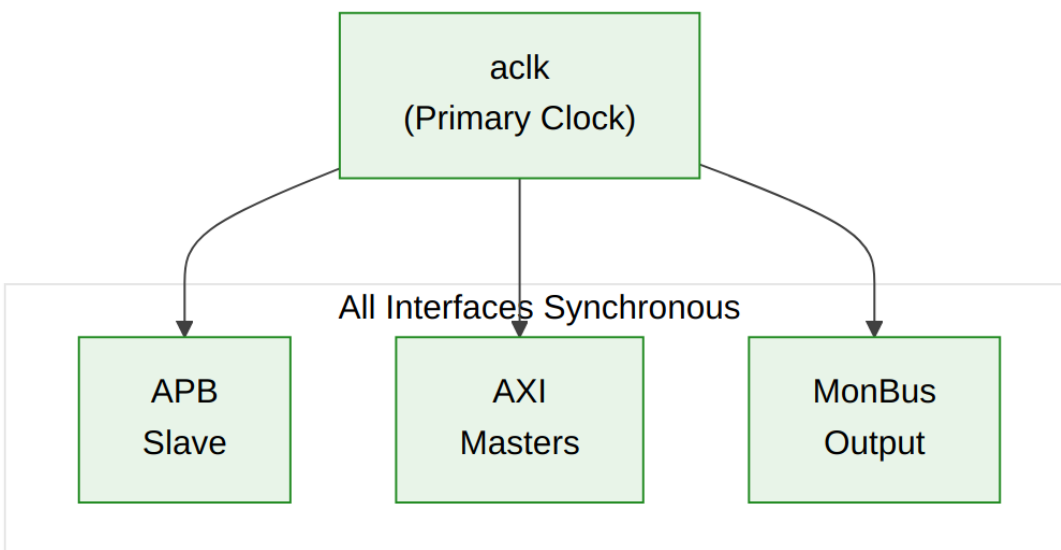
19.1.1 Primary Clock (aclk)

Parameter	Requirement
Frequency	100 - 400 MHz (target dependent)
Duty Cycle	40% - 60%
Jitter	<100 ps peak-to-peak
Domain	Single clock domain for all

Parameter	Requirement
	interfaces

19.1.2 Clock Relationship

All STREAM interfaces are synchronous to `aclk`:



Clock Distribution

Source: [06_clock_distribution.mmd](#)

[INTEG-1] External interfaces must be synchronous to `aclk`. Clock domain crossing is the system integrator's responsibility.

19.2 Reset Requirements

19.2.1 Reset Signal (`aresetn`)

Parameter	Requirement
Polarity	Active-low
Type	Asynchronous assert, synchronous deassert
Minimum Pulse	2 clock cycles
Glitch Filter	Not required (external synchronizer assumed)

19.2.2 Reset Sequence

Reset is asserted asynchronously and de-asserted synchronously:

1. `aresetn` goes low (asynchronous assertion)
2. Reset remains active for minimum 2 clock cycles
3. `aresetn` goes high on rising edge of `aclk` (synchronous release)
4. All blocks begin normal operation

19.2.3 Reset Behavior

Block	Reset State
APB Config	All registers to default
Scheduler	All channels IDLE
Descriptor Engine	FSM to IDLE, FIFOs empty
AXI Engines	No outstanding transactions
SRAM	Contents undefined
MonBus	FIFO empty, no output

19.3 Power Management

19.3.1 Clock Gating Support

STREAM supports external clock gating when idle:

Signal	Purpose
<code>all_channels_idle</code>	Indicates safe to gate clock

Clock Gating Sequence: 1. Check `STATUS.BUSY` = 0 2. Disable clock via external gate 3. Re-enable clock before any APB access

19.3.2 Low Power Recommendations

Technique	Implementation
Clock gating	Gate <code>aclk</code> when <code>all_channels_idle</code>
Power gating	Not recommended (state loss)
Voltage scaling	Supported if timing met

19.4 Timing Constraints

19.4.1 Clock Period Constraints

Example Vivado XDC constraints

```
create_clock -period 5.000 -name aclk [get_ports aclk]
```

APB interface (same clock)

```
set_input_delay -clock aclk 1.0 [get_ports s_apb_*]  
set_output_delay -clock aclk 1.0 [get_ports s_apb_pready]  
set_output_delay -clock aclk 1.0 [get_ports s_apb_prdata*]  
set_output_delay -clock aclk 1.0 [get_ports s_apb_pslverr]
```

AXI interfaces (same clock)

```
set_input_delay -clock aclk 1.0 [get_ports m_axi_*ready]  
set_input_delay -clock aclk 1.0 [get_ports m_axi_*rdata*]  
set_input_delay -clock aclk 1.0 [get_ports m_axi_*rresp*]  
set_input_delay -clock aclk 1.0 [get_ports m_axi_*bresp*]  
set_output_delay -clock aclk 1.0 [get_ports m_axi_*valid]  
set_output_delay -clock aclk 1.0 [get_ports m_axi_*addr*]  
set_output_delay -clock aclk 1.0 [get_ports m_axi_*data*]
```

19.4.2 False Path Constraints

Reset is asynchronous

```
set_false_path -from [get_ports aresetn]
```

Configuration registers to datapath (multi-cycle)

```
set_multicycle_path 2 -setup -from [get_cells *cfg_*]  
set_multicycle_path 1 -hold -from [get_cells *cfg_*]
```

19.5 CDC Considerations

19.5.1 Single Clock Domain

STREAM operates in a single clock domain. All clock domain crossing must be handled externally:

Interface	CDC Responsibility
APB Slave	External if APB on different clock
AXI Masters	External if AXI interconnect on different clock
MonBus	External if consumer on

Interface	CDC Responsibility
	different clock

19.5.2 CDC Recommendations

For systems requiring CDC:

1. Use async FIFO for MonBus output
2. Use AXI clock converter IP for AXI interfaces
3. Use APB bridge with clock crossing for APB

Last Updated: 2026-01-03

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20 Verification Strategy

20.1 Verification Approach

20.1.1 Multi-Level Verification

STREAM verification follows a hierarchical approach:

Level	Focus	Coverage Target
Block	Individual module function	100% line/toggle
Integration	Block-to-block interfaces	95% functional
System	End-to-end operation	90% use cases

20.2 Block-Level Verification

20.2.1 Test Categories

Block	Key Test Scenarios
APB Config	Register access, kick-off timing, error response
Descriptor Engine	Fetch, parse, chain, error handling
Scheduler	State transitions, multi-channel coordination
Channel Arbiter	Priority, fairness, starvation prevention
AXI Read Engine	Burst generation, error handling, backpressure
AXI Write Engine	Burst generation, response handling
SRAM Buffer	Read/write concurrent, full/empty
MonBus Reporter	Event generation, backpressure, overflow

20.2.2 Coverage Metrics

Metric	Target
Line coverage	100%
Toggle coverage	95%
FSM state coverage	100%
FSM transition coverage	100%
Parameter coverage	All supported values

20.3 Integration-Level Verification

20.3.1 Interface Verification

Interface	Test Focus
APB to Descriptor Engine	Kick-off handshake, blocking

Interface	Test Focus
	behavior
Descriptor Engine to Scheduler	Descriptor passing, backpressure
Scheduler to AXI Engines	Grant/request, data flow
AXI Engines to SRAM	Concurrent access, pointer management
All blocks to MonBus	Event generation, arbitration

20.3.2 Multi-Channel Scenarios

Scenario	Description
Sequential	Channels operate one at a time
Concurrent	All 8 channels active simultaneously
Priority	High priority preempts low priority
Fairness	Equal priority channels share fairly
Error isolation	Error in one channel, others continue

20.4 System-Level Verification

20.4.1 End-to-End Test Cases

Test Case	Description
Basic Transfer	Single descriptor, single channel
Chained Transfer	Multiple descriptors, single channel
Multi-Channel	Multiple channels, independent transfers
Scatter	Single source to multiple destinations
Gather	Multiple sources to single

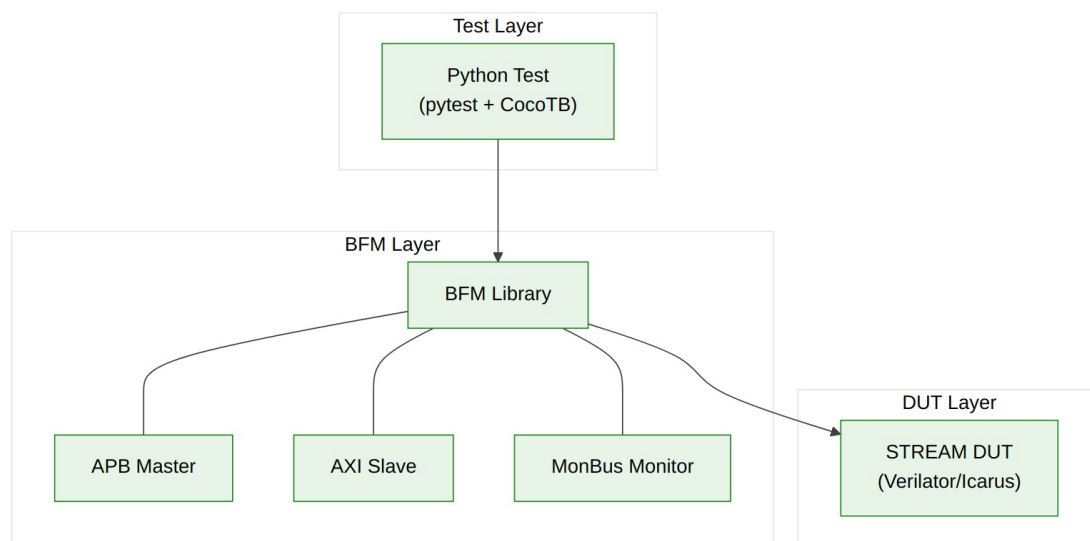
Test Case	Description
	destination
Maximum Throughput	Sustained bandwidth test
Error Injection	AXI errors, invalid descriptors
Stress	Random traffic, long duration

20.4.2 Performance Verification

Metric	Verification Method
Throughput	Measure bytes/cycle over large transfer
Latency	Measure kick-off to completion cycles
Efficiency	Compare actual vs theoretical maximum

20.5 Testbench Architecture

20.5.1 CocoTB-Based Verification



Testbench Architecture

Source: [07_testbench_stack.mmd](#)

20.5.2 Key Testbench Components

Component	Purpose
APB Master BFM	Drive configuration interface
AXI Slave BFM	Model memory system
MonBus Monitor	Capture and verify events
Scoreboard	Verify data integrity
Coverage Collector	Track functional coverage

20.6 Verification Environment

20.6.1 Simulator Support

Simulator	Status
Verilator	Primary (open-source)
Icarus Verilog	Secondary
Commercial (VCS, Questa)	Supported

20.6.2 Test Execution

Run all STREAM tests

```
pytest projects/components/stream/dv/tests/ -v
```

Run specific test category

```
pytest projects/components/stream/dv/tests/fub_tests/ -v
```

```
pytest projects/components/stream/dv/tests/integration_tests/ -v
```

Generate coverage report

```
pytest projects/components/stream/dv/tests/ --cov-report=html
```

20.7 Formal Verification

20.7.1 Properties for Formal

Category	Properties
Protocol	AXI handshake rules, APB timing
Liveness	No deadlock in arbitration

Category	Properties
Safety	No data corruption, proper error handling
Functional	Descriptor chaining correctness

20.7.2 Formal Tool Usage

Recommended for: - AXI protocol compliance - Arbiter deadlock freedom - FSM reachability analysis

20.8 Sign-Off Criteria

20.8.1 Block-Level Sign-Off

- ☐ 100% line coverage
- ☐ 95% toggle coverage
- ☐ All FSM states exercised
- ☐ All FSM transitions exercised
- ☐ No simulation errors
- ☐ No lint warnings

20.8.2 Integration Sign-Off

- ☐ All interface scenarios passed
- ☐ Multi-channel concurrency verified
- ☐ Error handling verified
- ☐ Performance targets met

20.8.3 System Sign-Off

- ☐ All end-to-end test cases passed
 - ☐ Throughput targets achieved
 - ☐ Latency targets achieved
 - ☐ 72-hour stress test passed
-

Last Updated: 2026-01-03

21 Product Requirements Document (PRD)

21.1 STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory

Version: 1.0 **Date:** 2025-10-17 **Status:** Nearly Complete - Final Integration Pending **Owner:** RTL Design Sherpa Project **Parent Document:** /PRD.md

21.2 1. Executive Summary

The **STREAM** (Scatter-gather Transfer Rapid Engine for AXI Memory) is a simplified DMA-style accelerator designed for efficient memory-to-memory data movement with descriptor-based scatter-gather support. STREAM serves as a beginner-friendly tutorial demonstrating descriptor-based DMA engine design patterns while maintaining production-quality RTL practices.

21.2.1 1.1 Quick Stats

- **Modules:** ~8-10 SystemVerilog files (estimated)
- **Channels:** Maximum 8 independent channels
- **Interfaces:** APB (config), AXI4 (descriptor fetch + data read/write), MonBus (monitoring)
- **Architecture:** Simplified from RAPIDS - pure memory-to-memory
- **Tutorial Focus:** Aligned addresses only, straightforward data flow
- **Status:** Nearly complete - config interface and top-level wrapper pending

21.2.2 1.2 Project Goals

- **Primary:** Educational DMA engine demonstrating scatter-gather descriptor chains
 - **Secondary:** Production-quality RTL suitable for FPGA/ASIC implementation
 - **Tertiary:** Foundation for understanding more complex DMA architectures (e.g., RAPIDS)
-

21.3 2. Key Design Principles

21.3.1 2.1 Simplifications from RAPIDS

STREAM is intentionally simplified for tutorial purposes:

Feature	RAPIDS	STREAM
Network Interfaces	Yes (Network master/slave)	✗ No (pure memory-to-memory)
Address Alignment	Complex fixup logic	✓ Aligned addresses only
Credit Management	Exponential encoding	✓ Simple transaction limits
Control Engines	Control read/write engines	✗ No (direct APB config)
Descriptor Length	Chunks (4-byte)	✓ Beats (data width)
Program Engine	Complex alignment FSM	✓ Simplified coordination

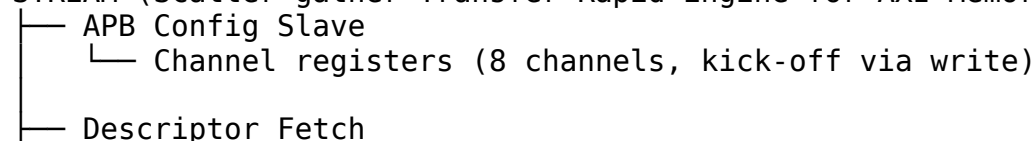
21.3.2 2.2 Tutorial-Friendly Features

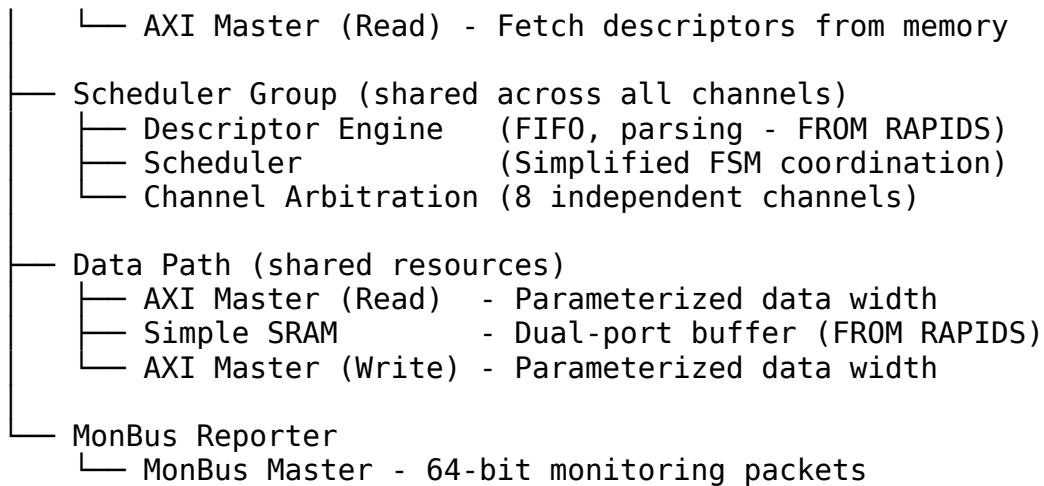
- **Aligned Addresses:** Source and destination addresses must be aligned to data width
- **Length in Beats:** Descriptor length specified in data beats (not bytes/chunks)
- **Single APB Write:** One APB register write kicks off entire descriptor chain
- **No Circular Buffers:** Chained descriptors with explicit termination
- **Parameterized Engines:** Multiple AXI engine versions (compile-time selection)

21.4 3. Architecture Overview

21.4.1 3.1 Top-Level Block Diagram

STREAM (Scatter-gather Transfer Rapid Engine for AXI Memory)





21.4.23.2 Data Flow

Descriptor-Based Transfer Sequence:

1. Software writes to APB channel register
↓
2. Descriptor fetch via AXI descriptor master
↓
3. Descriptor Engine parses descriptor fields
↓
4. Scheduler coordinates data transfer:
 - a. AXI Read Engine fetches source data → SRAM buffer
 - b. AXI Write Engine writes SRAM → destination
 ↓
5. Check for chained descriptor (next_descriptor_ptr != 0)
↓ (if chained)
6. Fetch next descriptor, repeat from step 3
↓ (if last)
7. Generate MonBus completion packet

Channel Independence: - 8 channels operate independently - All channels share: SRAM, AXI data masters, descriptor fetch master - Arbitration required for shared resources

21.5 4. Interfaces

21.5.14.1 External Interfaces

Interface	Type	Width	Purpose	Notes
APB Slave	Slave	32-bit	Configuration, channel kick-off	Write to channel register

Interface	Type	Width	Purpose	Notes
				starts transfer
AXI Master (Descriptor)	Master	256-bit	Fetch descriptors from memory	Dedicated descriptor fetch path
AXI Master (Data Read)	Master	Parameterizable	Read source data	Multiple engine versions (compile-time)
AXI Master (Data Write)	Master	Parameterizable	Write destination data	Multiple engine versions (compile-time)
MonBus Master	Master	64-bit	Monitor packet output	Standard AMBA format

21.5.24.2 Descriptor Format

256-bit Descriptor Structure:

Bits	Field	Description
[63:0]	src_addr	Source address (64-bit, must be aligned to data width)
[127:64]	dst_addr	Destination address (64-bit, must be aligned to data width)
[159:128]	length	Transfer length in BEATS (not bytes!)
[191:160]	next_descriptor_ptr	Address of next descriptor (0 = last in chain)
[192]	valid	Descriptor is valid
[193]	interrupt	Generate interrupt on completion
[194]	last	Last descriptor in chain (explicit flag)
[195]	error	Error status (used for reporting)

Bits	Field	Description
[199:196]	channel_id	Channel ID (0-7)
[207:200]	priority	Transfer priority (for arbitration)
[255:208]	reserved	Reserved for future use

Key Descriptor Features: - ✓ **Chained descriptors:** next_descriptor_ptr links to next descriptor - ✗ **No circular buffers:** Explicit termination (last flag or ptr=0) - ✓ **Length in beats:** Simplified for tutorial (no byte/chunk conversion) - ✓ **Aligned addresses:** Tutorial constraint (performance hidden for now)

21.6 5. Key Components

21.6.15.1 Descriptor Engine (APB-Only for STREAM)

Source: Adapted from RAPIDS descriptor_engine.sv

Purpose: - Autonomous descriptor fetch and chaining - APB interface for initial descriptor address - AXI read interface for descriptor memory fetches - Descriptor FIFO storage and distribution

Key Features: - ✓ **Autonomous chaining:** Automatically fetches next descriptor if next_descriptor_ptr != 0 AND last == 0 - ✓ **Address validation:** Validates next descriptor addresses against cfg_addr0/1_base/limit - ✓ **APB blocking:** APB blocked until channel_idle == 1 (channel fully idle) - ✓ **Error handling:** AXI errors stop chaining, set descriptor_error, block descriptor_valid

Adaptations from RAPIDS: - ✓ **APB-only:** Single APB write kicks off entire descriptor chain - ✓ **Descriptor Read Address FIFO:** 2-deep FIFO stores addresses for AXI fetch (APB + chaining) - ✓ **Chaining logic:** Descriptor engine autonomously manages next_descriptor_ptr chaining

Idle Signal: - descriptor_engine_idle asserted when: - FSM in RD_IDLE state - No pending descriptor fetches (address FIFO empty) - No active AXI transactions

21.6.25.2 Scheduler Group (Integration Wrapper)

Purpose: Wraps descriptor engine and scheduler into a single channel processing unit

Architecture:

```
scheduler_group (
    // APB interface (from APB config slave)
    .apb_valid      (apb_valid),
    .apb_ready      (apb_ready),    // Blocked when channel not idle
    .apb_addr       (descriptor_addr),
```

```

// Channel idle signal composition (CRITICAL!)
.channel_idle      (channel_idle),

// Descriptor → Scheduler flow
.descriptor_valid  (desc_valid),
.descriptor_ready  (desc_ready),
.descriptor_packet (desc_packet),

// Data engine interfaces
.datard_*          (datard_*),    // Read engine
.datawr_*          (datawr_*),    // Write engine

// Status
.scheduler_idle    (sched_idle),
.descriptor_idle    (desc_idle)
);

```

Channel Idle Signal Composition:

```

// Channel is idle ONLY when BOTH sub-blocks are idle
assign channel_idle = scheduler_idle && descriptor_engine_idle;

```

Why Both Signals Matter:

Signal	Indicates	Used For
scheduler_idle	No active data transfers, all descriptors processed	Prevents new APB request during active transfer
descriptor_engine_idle	No pending descriptor fetches (FIFO empty)	Prevents new APB request during chaining
channel_idle (AND of both)	Channel fully quiescent	Gates APB interface

APB Blocking Logic:

```

// Descriptor engine blocks APB when channel not idle
assign apb_ready = apb_skid_ready_in &&
                  !r_channel_reset_active &&
                  w_desc_addr_fifo_empty &&
                  channel_idle;
// No pending fetches
// Scheduler +
descriptor_idle

```

Example Scenario:

1. Software writes APB → descriptor_addr = 0x1000
 - channel_idle = 1 (both idle)
 - APB accepted

2. Descriptor engine fetches descriptor @ 0x1000
 - descriptor_engine_idle = 0 (fetch in progress)
 - channel_idle = 0
 - APB BLOCKED
3. Descriptor pushed to scheduler
 - descriptor_engine_idle = 1 (fetch complete)
 - scheduler_idle = 0 (transfer starting)
 - channel_idle = 0
 - APB BLOCKED
4. Scheduler completes data transfer
 - Descriptor has next_descriptor_ptr = 0x1100 (chained!)
 - Descriptor engine autonomously fetches @ 0x1100
 - descriptor_engine_idle = 0 (autonomous fetch)
 - channel_idle = 0
 - APB BLOCKED
5. Final descriptor completes (last = 1 OR next_ptr = 0)
 - scheduler_idle = 1 (transfer done)
 - descriptor_engine_idle = 1 (no more fetches)
 - channel_idle = 1
 - APB UNBLOCKED (ready for next transfer!)

Key Insight: The AND gate ensures software cannot interrupt a descriptor chain in progress!

21.6.35.3 Scheduler (Simplified from RAPIDS)

Purpose: - Coordinate descriptor-to-data-transfer flow - Manage 8 independent channels - Arbitrate shared resources (SRAM, AXI masters)

FSM States:

```
typedef enum logic [7:0] {
    SCHED_IDLE = 8'b00000001, // Idle, waiting for
channel activation
    SCHED_FETCH_DESCRIPTOR = 8'b00000010, // Fetch descriptor via
AXI master
    SCHED_PARSE_DESCRIPTOR = 8'b00000100, // Parse descriptor fields
    SCHED_READ_PHASE = 8'b00001000, // Coordinate read engine
    SCHED_WRITE_PHASE = 8'b00010000, // Coordinate write engine
    SCHED_CHAIN_CHECK = 8'b00100000, // Check for next
descriptor
    SCHED_COMPLETE = 8'b01000000, // Transfer complete,
report status
    SCHED_ERROR = 8'b10000000 // Error state
} scheduler_state_t;
```

Key Differences from RAPIDS: - ✗ No credit management (just simple transaction limits) - ✗ No program engine coordination (no alignment fixup) - ✓ Simplified FSM (no control read/write phases)

21.6.45.3 AXI Read Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming reads from memory to SRAM buffer

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: FSMs are horrible for performance! Instead, use: - **Arbiter** selects which channel's `datard_*` interface gets access - **Streaming pipeline** continuously moves data when granted - **Data interface** (`datard_valid`, `datard_ready`, `datard_beats_remaining`) controls flow

Data Read Interface (per channel):

```
// Channel requests read access
input logic      datard_valid;           // Channel has read request
output logic     datard_ready;           // Engine ready for request
input logic [63:0] datard_addr;          // Source address (aligned)
input logic [31:0] datard_beats_remaining; // Beats left to read
input logic [7:0]  datard_burst_len;     // Preferred burst length
input logic [3:0]  datard_channel_id;    // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding read, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding reads, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

1. Arbiter grants channel access based on priority
2. Engine accepts `datard_*` request (continuous streaming)
3. AXI AR channel issues read burst
4. AXI R channel streams data → SRAM (no FSM stalls!)
5. Engine updates `beats_remaining`, accepts next request
6. Different channels can have different burst lengths (e.g., CH0: 8 beats, CH1: 16 beats)

Example: Different Burst Lengths per Channel

```
// Channel 0 prefers 8-beat bursts
datard_burst_len[0] = 8'd8;

// Channel 1 prefers 16-beat bursts
datard_burst_len[1] = 8'd16;

// Engine adapts to requested burst length (within MAX_BURST_LEN)
```

21.6.55.4 AXI Write Engine (Streaming Pipeline - NO FSM)

Purpose: High-performance streaming writes from SRAM buffer to memory

Architecture: Pipelined streaming design (NO FSM for performance)

Key Insight: Same as read engine - no FSMs! Use streaming pipeline with arbiter.

Data Write Interface (per channel):

```
// Channel requests write access
input logic      datawr_valid;           // Channel has write
request                                                  request
output logic     datawr_ready;           // Engine ready for
request                                                  request
input logic [63:0] datawr_addr;           // Destination address
(aligned)
input logic [31:0] datawr_beats_remaining; // Beats left to write
input logic [7:0]  datawr_burst_len;      // Preferred burst length
input logic [3:0]  datawr_channel_id;     // Channel ID for tracking
```

Multiple Versions (Compile-Time Selection): 1. **Version 1 - Basic:** Single outstanding write, fixed burst length 2. **Version 2 - Pipelined:** Multiple outstanding writes, configurable bursts 3. **Version 3 - Adaptive:** Dynamic burst sizing based on remaining beats

Operation:

1. Arbiter grants channel access based on priority
2. Engine accepts datawr_* request (continuous streaming)
3. Engine reads data from SRAM
4. AXI AW channel issues write address
5. AXI W channel streams data (no FSM stalls!)
6. AXI B channel receives response, updates beats_remaining
7. Different channels can have different burst lengths (e.g., CH0: 8 beats, CH2: 32 beats)

Read/Write Asymmetry Example:

```
// Channel can use different burst lengths for read vs write
// Example: Read in small bursts, write in large bursts
datard_burst_len[0] = 8'd8; // Read: 8 beats
datawr_burst_len[0] = 8'd16; // Write: 16 beats

// Engine handles asymmetry via SRAM buffering
```

21.6.65.5 Simple SRAM

Source: Direct copy from RAPIDS simple_sram.sv

Purpose: - Dual-port SRAM buffer - Decouples read and write engines - Shared across all channels (arbitration required)

Why Reuse: - Standard dual-port SRAM design - Proven in RAPIDS integration tests - Parameterizable depth and width

21.7 6. Configuration and Control

21.7.16.1 APB Register Map

Offset	Register	Access	Description
0x0000	GLOBAL_CTRL	RW	Global enable, reset
0x0004	GLOBAL_STATUS	RO	Global status, error flags
0x0100	CH0_CTRL	WO	Channel 0 kick-off (write descriptor address)
0x0104	CH0_STATUS	RO	Channel 0 status
0x0108	CH0_DESC_ADDR	RO	Channel 0 current descriptor address
0x010C	CH0_BYTES_XFER	RO	Channel 0 bytes transferred
...	...	...	... (repeat for channels 1-7)
0x0200	CH1_CTRL	WO	Channel 1 kick-off
...			
0x0700	CH7_CTRL	WO	Channel 7 kick-off

Kick-Off Sequence: 1. Software writes descriptor address to CHx_CTRL register 2. STREAM fetches descriptor from memory 3. Transfer begins automatically 4. If chained, STREAM follows next_descriptor_ptr automatically 5. Completion reported via MonBus packet

21.7.26.2 Channel Configuration

Each channel independently configurable: - **Descriptor start address:** Written to CHx_CTRL - **Priority:** Encoded in descriptor (arbitration) - **Interrupt enable:** Per-descriptor flag - **Status monitoring:** Read CHx_STATUS

21.8 7. Resource Sharing and Arbitration

21.8.17.1 Shared Resources

All channels share: 1. **Descriptor Fetch AXI Master** - Fetches descriptors for all channels 2. **Data Read AXI Master** - Reads source data for all channels 3. **Data Write AXI Master** - Writes destination data for all channels 4. **SRAM Buffer** - Shared buffer (dual-port, but still arbitrated) 5. **MonBus Reporter** - Single monitor output

21.8.27.2 Arbitration Strategy

Priority-Based Round-Robin: - Channels have priority field in descriptor - Higher priority = serviced first - Within same priority: round-robin - Prevents starvation with timeout

Example Arbitration:

Channel 0: Priority 7 (highest)
Channel 1: Priority 5
Channel 2: Priority 5
Channel 3: Priority 3

Service order: CH0 → CH1 → CH2 (round-robin) → CH0 → CH1 → CH2 → CH3 ...

21.9 8. Error Detection and Recovery

21.9.18.1 Error Types

Error Type	Detection	Response
Invalid descriptor	Valid bit = 0	Skip, move to next
Alignment error	Address not aligned	Set error flag, halt channel
AXI SLVERR	AXI response	Set error flag, halt channel
AXI DECERR	AXI response	Set error flag, halt channel
Timeout	Transaction timeout	Set error flag, halt channel
SRAM overflow	Buffer full	Backpressure, wait

21.9.28.2 Error Recovery

Per-Channel Error Handling: - Error sets channel to CH_ERROR state - Channel halts, does not affect other channels - Software must: 1. Read CHx_STATUS to identify error 2. Clear error condition 3. Re-kick channel with new descriptor

No Automatic Retry: - Tutorial design keeps error handling simple - Software responsible for retry logic

21.10 9. MonBus Integration

21.10.1 9.1 Standard MonBus Format

Uses standard 64-bit MonBus packet format: - [63:60] Packet Type (0=ERROR, 1=COMPL, etc.) - [59:57] Protocol (custom STREAM protocol) - [56:53] Event Code (STREAM-specific events) - [52:47] Channel ID (0-7) - [46:43] Unit ID (unused for STREAM) - [42:35] Agent ID (unused for STREAM) - [34:0] Event Data (address, byte count, etc.)

21.10.2 9.2 STREAM Event Codes

Code	Event	Description
0x0	DESC_START	Descriptor started
0x1	DESC_COMPLETE	Descriptor completed
0x2	READ_START	Read phase started
0x3	READ_COMPLETE	Read phase completed
0x4	WRITE_START	Write phase started
0x5	WRITE_COMPLETE	Write phase completed
0x6	CHAIN_FETCH	Chained descriptor fetch
0xF	ERROR	Error occurred

21.10.3 9.3 Default Configuration

Tutorial-Friendly Defaults: - **Errors only:** `cfg_error_enable = 1`, all others = 0 - **Interrupts:** Descriptor flag controls per-transfer interrupt - **Reduces MonBus traffic** for beginner understanding

21.11 10. Design Constraints

21.11.1 10.1 Tutorial Constraints (Intentional Simplifications)

Constraint	Rationale
Aligned addresses only	Simplify data path, hide alignment complexity
Length in beats	Avoid byte/chunk conversion math
No circular buffers	Explicit termination easier to understand
Single APB kick-off	Simple software model
No credit management	Avoid exponential encoding complexity

21.11.2 10.2 Implementation Constraints

Parameter	Value	Notes
Max channels	8	Compile-time parameter
Max burst length	256	AXI4 spec limit
Descriptor size	256 bits	4 × 64-bit words
SRAM depth	Parameterizable	Typical: 1024-4096 entries
Data width	Parameterizable	Typical: 512-bit

21.12 11. Verification Strategy

21.12.1 11.1 Test Organization

```
projects/components/stream/dv/tests/
├── fub_tests/                # Functional Unit Block tests
│   ├── descriptor_engine/    # Copy from RAPIDS (adapt imports)
│   ├── scheduler/           # Simplified scheduler tests
│   ├── axi_engines/         # Read/write engine tests
│   └── sram/                 # SRAM tests
└── integration_tests/       # Multi-block scenarios
    └── single_channel/      # Single channel transfers
```

└─ multi_channel/	# 8-channel concurrent
└─ chained_descriptors/	# Descriptor chain tests
└─ error_handling/	# Error recovery tests

21.12.2 11.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

21.13 12. Performance Characteristics

21.13.1 12.1 Throughput by Engine Version

V1 (Low Performance - Tutorial Mode): - **Throughput:** 0.14 beats/cycle (DDR4), 0.40 beats/cycle (SRAM) - **Architecture:** Single outstanding transaction, blocks on completion - **Use Case:** Tutorial examples, embedded systems, low-latency SRAM

V2 (Medium Performance - Command Pipelined): - **Throughput:** 0.94 beats/cycle (DDR4), 0.85 beats/cycle (SRAM) - **Improvement:** 6.7x over V1 (DDR4), 2.1x over V1 (SRAM) - **Architecture:** Command queue (4-8 deep), hides memory latency - **Use Case:** General-purpose FPGA, DDR3/DDR4, best area efficiency

V3 (High Performance - Out-of-Order): - **Throughput:** 0.98 beats/cycle (DDR4), 0.92 beats/cycle (SRAM) - **Improvement:** 7.0x over V1 (DDR4), 2.3x over V1 (SRAM) - **Architecture:** OOO command selection, maximizes memory controller efficiency - **Use Case:** Datacenter, ASIC, HBM2, high-performance memory

Key Insight: Benefit scales with memory latency. V1 throughput degrades from 40% (SRAM) to 14% (DDR4), while V2/V3 maintain 94-98% throughput regardless of latency.

Configuration Parameters: - `ENABLE_CMD_PIPELINE = 0`: V1 (default, tutorial mode) - `ENABLE_CMD_PIPELINE = 1`: V2 (command pipelined) - `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read): V3

Factors Affecting Throughput: - Memory latency (V2/V3 hide latency via pipelining) - SRAM buffer size (limits burst pipelining) - Channel arbitration overhead - Descriptor fetch latency - Engine version (V1/V2/V3 configuration)

21.13.2 12.2 Latency

Transfer Latency Breakdown: - Descriptor fetch: ~10-50 cycles (depends on memory) - Read phase: $(\text{length} / \text{burst_len}) \times \text{burst_latency}$ - Write phase: $(\text{length} / \text{burst_len}) \times \text{burst_latency}$ - End-to-end: Typically <200 cycles for small transfers (V1), <100 cycles (V2/V3 pipelined)

V2/V3 Latency Hiding: - Command pipelining overlaps descriptor fetch, read, write operations - Multiple outstanding transactions hide memory latency - OOO completion (V3) reduces head-of-line blocking

21.13.3 12.3 Resource Utilization by Engine Version

V1 Configuration (Tutorial - Minimum Area): - Total: ~9,500 LUTs + 64 KB SRAM - Descriptor Engine (8x): ~2,400 LUTs - Scheduler (8x): ~3,200 LUTs - AXI Read Engine: ~1,250 LUTs - AXI Write Engine: ~1,250 LUTs - SRAM Controller: ~1,600 LUTs - APB Config: ~350 LUTs - MonBus AXIL Group: ~1,000 LUTs

V2 Configuration (Balanced - Best Area Efficiency): - Total: ~11,000 LUTs + 64 KB SRAM (1.16x area) - AXI Read Engine: ~2,000 LUTs (1.6x increase, 6.7x throughput) - AXI Write Engine: ~2,500 LUTs (2.0x increase, 6.7x throughput) - Other blocks: Same as V1

V3 Configuration (Maximum Performance): - Total: ~14,000 LUTs + 64 KB SRAM (1.47x area) - AXI Read Engine: ~3,500 LUTs (2.8x increase, 7.0x throughput) - AXI Write Engine: ~4,000 LUTs (3.2x increase, 7.0x throughput) - Other blocks: Same as V1

Area Efficiency Comparison: - V1: 1.00 throughput / 1.00 area = 1.00 - V2: 6.70 throughput / 1.16 area = 5.78 (best efficiency) - V3: 7.00 throughput / 1.47 area = 4.76

Recommendation: V2 provides best area efficiency for most use cases. V3 justified only for high-performance memory controllers that support OOO responses.

21.14 13. Development Roadmap

21.14.1 13.1 Phase 1: Foundation (Current)

- ✓ Directory structure
- ✓ Package definitions (`stream_pkg.sv`)
- ✓ Imports header (`stream_imports.svh`)
- 🕒 Documentation (this PRD)

21.14.2 13.2 Phase 2: Core Blocks

- Adapt descriptor engine from RAPIDS
- Design simplified scheduler

- Create APB config interface
- Copy simple SRAM from RAPIDS

21.14.3 13.3 Phase 3: Data Path

- AXI read engine (version 1 - basic)
- AXI write engine (version 1 - basic)
- SRAM integration
- Channel arbitration

21.14.4 13.4 Phase 4: Integration

- Top-level module
- MonBus reporter
- Integration testbench
- Single-channel validation

21.14.5 13.5 Phase 5: Multi-Channel

- 8-channel support
- Arbiter implementation
- Multi-channel tests
- Performance tuning

21.14.6 13.6 Phase 6: Advanced Engines (Future - V2/V3)

Goal: Add parameterized high-performance engine variants

V2 - Command Pipelined (Medium Performance): - Command queue implementation (4-8 deep) - W drain FSM for write engine - B response scoreboard (write) or in-order R reception (read) - Per-command SRAM pointer tracking - Parameter: `ENABLE_CMD_PIPELINE = 1` - Expected: 6.7x throughput improvement over V1

V3 - Out-of-Order Completion (High Performance): - OOO command selection logic - Transaction ID matching (AXI ID to queue entry) - SRAM data availability checking (write engine) - R beat matching to queue entry (read engine) - Parameters: `ENABLE_CMD_PIPELINE = 1`, `ENABLE_OOO_DRAIN = 1` (write) or `ENABLE_OOO_READ = 1` (read) - Expected: 7.0x throughput improvement over V1

Deliverables: - Updated RTL with parameterization - Performance comparison tests (V1 vs V2 vs V3) - Tutorial documentation explaining trade-offs - Area/throughput measurements on target FPGA

21.15 14. Educational Value

21.15.1 14.1 Learning Objectives

STREAM teaches: 1. **Descriptor-based DMA design patterns** - Descriptor structure and parsing - Chained descriptors (scatter-gather) - Descriptor fetch via AXI

2. **AXI4 Memory Interface Usage**

- Burst transactions
- Address/data/response channels
- Outstanding transactions

3. **Resource Sharing and Arbitration**

- Multiple channels sharing AXI masters
- Priority-based arbitration
- Conflict resolution

4. **FSM Design and Coordination**

- Multiple interconnected FSMs
- State machine composition
- Error handling

5. **Buffer Management**

- SRAM-based buffering
- Rate matching
- Flow control

21.15.2 14.2 Progression Path

Learning Sequence: 1. **STREAM (this project):** Memory-to-memory DMA, aligned addresses 2.

STREAM Extended: Add alignment fixup, more complex scenarios 3. **RAPIDS:** Add network interfaces, credit management, full complexity

21.16 15. Success Criteria

21.16.1 15.1 Functional

- ✓ Single descriptor transfer working
- ✓ Chained descriptors (2-3 deep)
- ✓ Multi-channel operation (8 channels concurrent)
- ✓ Error detection and reporting

- ✓ MonBus packet generation
- ✓ >90% functional test coverage

21.16.2 15.2 Quality

- ✓ Verilator compiles with 0 warnings
- ✓ All tests passing (100% success rate)
- ✓ Comprehensive documentation
- ✓ Tutorial-quality code comments

21.16.3 15.3 Performance

- ✓ Achieves >80% of theoretical AXI bandwidth
 - ✓ <5K LUT utilization (excluding SRAM)
 - ✓ <200 cycle end-to-end latency for small transfers
-

21.17 16. Open Questions (For Review)

21.17.1 16.1 Descriptor Engine Adaptation

- **Q:** Should descriptor engine use APB-only, or support additional modes?
- **A (pending):** TBD - depends on software use case preference

21.17.2 16.2 AXI Descriptor Master

- **Q:** Fixed 256-bit width, or parameterizable?
- **A (pending):** Propose fixed 256-bit for simplicity

21.17.3 16.3 Channel Arbitration

- **Q:** Fixed priority, or dynamic priority based on age/fairness?
- **A (pending):** Propose fixed priority with round-robin for tutorial simplicity

21.17.4 16.4 SRAM Partitioning

- **Q:** Single shared SRAM, or per-channel SRAMs?
 - **A (pending):** Propose single shared SRAM with arbitration (matches RAPIDS pattern)
-

21.18 16. Attribution and Contribution Guidelines

21.18.1 16.1 Git Commit Attribution

When creating git commits for STREAM documentation or implementation:

Use:

Documentation and implementation support by Claude.

Do NOT use:

Co-Authored-By: Claude <noreply@anthropic.com>

Rationale: STREAM documentation and organization receives AI assistance for structure and clarity, while design concepts and architectural decisions remain human-authored.

21.19 16.2 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

/mnt/data/github/rtldesignsherpa/projects/components/stream/docs/

Quick Command: Use the provided shell script:

```
cd /mnt/data/github/rtldesignsherpa/projects/components/stream/docs
./generate_pdf.sh
```

The shell script will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_spec index file 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

 **See:** bin/md_to_docx.py for complete implementation details

21.20 17. References

21.20.1 17.1 Internal Documentation

- **RAPIDS PRD:** projects/components/rapids/PRD.md - Parent architecture
- **RAPIDS Descriptor Engine:**
projects/components/rapids/rtl/rapids_fub/descriptor_engine.sv
- **RAPIDS Simple SRAM:**
projects/components/rapids/rtl/rapids_fub/simple_sram.sv
- **AMBA PRD:** rtl/amba/PRD.md - MonBus integration

- **Repository Guide:** /CLAUDE.md - Design patterns and conventions

21.20.2 17.2 External References

- **AXI4 Specification:** ARM IHI0022E
 - **APB Specification:** ARM IHI0024C
 - **CocoTB Documentation:** <https://docs.cocotb.org/>
 - **Verilator Manual:** <https://verilator.org/guide/latest/>
-

Document Version: 1.0 **Last Updated:** 2025-10-17 **Review Cycle:** Weekly during initial design
Next Review: TBD (after user feedback) **Owner:** RTL Design Sherpa Project

21.21 Navigation

- **← Back to Root:** /PRD.md
- **Parent Architecture:** projects/components/rapids/PRD.md
- **AI Guidance:** CLAUDE.md (to be created)
- **Quick Start:** README.md (to be created)

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22 Claude Code Guide: STREAM Subsystem

Version: 1.0 **Last Updated:** 2025-10-17 **Purpose:** AI-specific guidance for working with STREAM subsystem

22.1 Quick Context

What: STREAM - Scatter-gather Transfer Rapid Engine for AXI Memory **Status:** 🟡 Initial design - tutorial-focused DMA engine **Your Role:** Help users build a beginner-friendly descriptor-based DMA engine

📖 **Complete Specification:** projects/components/stream/PRD.md ← **Always reference this for technical details**

22.2 Global Requirements Reference

IMPORTANT: Review `/GLOBAL_REQUIREMENTS.md` for all mandatory requirements

All mandatory requirements are consolidated in the global requirements document: - **See:** `/GLOBAL_REQUIREMENTS.md` - Repository-wide mandatory requirements - **STREAM-Specific:** Attribution format, tutorial focus (intentional simplifications) - **Universal:** TB location, three methods, TBBase inheritance, 100% success

This CLAUDE.md provides STREAM-specific guidance. Also review: - Root `/CLAUDE.md` - Repository-wide patterns - `projects/components/CLAUDE.md` - Project area standards (reset macros, FPGA attributes) - `bin/TBClasses/CLAUDE.md` - Framework usage patterns

22.3 Critical Rules for This Subsystem

22.3.1 Rule #0: Attribution Format for Git Commits

IMPORTANT: When creating git commit messages for STREAM documentation or code:

Use:

Documentation and implementation support by Claude.

Do NOT use:

Co-Authored-By: Claude <noreply@anthropic.com>

Rationale: STREAM receives AI assistance for structure and clarity, while design concepts and architectural decisions remain human-authored.

22.3.2 Rule #0.1: TUTORIAL FOCUS - Intentional Simplifications

⚠️ **STREAM is INTENTIONALLY SIMPLIFIED for educational purposes** ⚠️

Key Simplifications (DO NOT “fix” these): 1. ✓ **Aligned addresses only** - No alignment fixup logic (kept for RAPIDS) 2. ✓ **Length in beats** - Not bytes or chunks (simplifies math) 3. ✓ **No circular buffers** - Explicit chain termination only 4. ✓ **No credit management** - Simple transaction limits 5. ✓ **Pure memory-to-memory** - No network interfaces

When users ask “Can we add alignment fixup?”: - ✓ **Correct answer:** “STREAM intentionally keeps addresses aligned for tutorial simplicity. For complex alignment, see RAPIDS.” - ✗ **Wrong answer:** “Sure, let me add alignment logic...” (defeats tutorial purpose!)

22.3.3 Rule #0.1: Testbench Location and Test Structure (MANDATORY)

 **See:** /GLOBAL_REQUIREMENTS.md Section 2.1 for complete requirement

STREAM-Specific Directory Structure:

```
projects/components/stream/dv/
├── tbclasses/                                # ★ STREAM TB classes (project
area!)
│   ├── scheduler_tb.py                      # Scheduler testbench
│   ├── descriptor_engine_tb.py             # Descriptor engine testbench
│   └── axi_engine_tb.py                    # AXI engine testbenches
└── tests/fub_tests/                         # Test runners import from
tbclasses/
```

STREAM Import Pattern:

```
# Import STREAM TB from project area
from projects.components.stream.dv.tbclasses.scheduler_tb import
StreamSchedulerTB
```

```
# Shared utilities from framework
from TBClasses.shared.tbbase import TBBBase
```

 **Complete Pattern:** projects/components/rapids/CLAUDE.md Rule #0.1

22.3.4 Rule #0.2: Three Mandatory TB Methods (MANDATORY)

 **See:** /GLOBAL_REQUIREMENTS.md Section 2.2 for complete requirement

STREAM-Specific Context:

STREAM has simpler config requirements than RAPIDS - most config can be set after reset.

Standard STREAM Pattern:

```
class StreamSchedulerTB(TBBBase):
    async def setup_clocks_and_reset(self):
        """Standard STREAM initialization"""
        await self.start_clock('clk', freq=10, units='ns')
        await self.assert_reset()
        await self.wait_clocks('clk', 10)
        await self.deassert_reset()
        await self.wait_clocks('clk', 5)

    async def assert_reset(self):
        self.dut.rst_n.value = 0

    async def deassert_reset(self):
        self.dut.rst_n.value = 1
```

Note: Unlike RAPIDS, STREAM typically doesn't need config set before reset.

22.3.5 Rule #1: REUSE from RAPIDS Where Appropriate

Direct Reuse (No Modification): - ✓ `descriptor_engine.sv` - Works with STREAM descriptors -
✓ `simple_sram.sv` - Standard dual-port SRAM - ✓ Descriptor engine tests - Adapt imports only

Adapt from RAPIDS: - ⚠ `scheduler.sv` - **Simplify FSM** (no credit management, no control engines) - ⚠ AXI engines - **Create simplified versions** (no alignment fixup)

Create New for STREAM: - 📝 APB config interface - PeakRDL-generated (like HPET), 8 channels, kick-off registers - 📝 Top-level integration - Different interface set

Always Ask Yourself: “Can I reuse from RAPIDS instead of creating new?”

22.3.6 Rule #2: Descriptor Format is DIFFERENT from RAPIDS

STREAM Descriptor (256-bit): - `src_addr` (64-bit), `dst_addr` (64-bit), `length` (**beats**), `next_descriptor_ptr` (32-bit) - **Length is in BEATS, not chunks!**

RAPIDS Descriptor: - Uses chunks (4-byte units) - Has alignment metadata

When comparing/referencing RAPIDS: - ✓ “RAPIDS uses chunks, STREAM uses beats for tutorial simplicity” - ✗ Don't assume RAPIDS descriptor format applies to STREAM

22.3.7 Rule #3: Know the Shared Resources

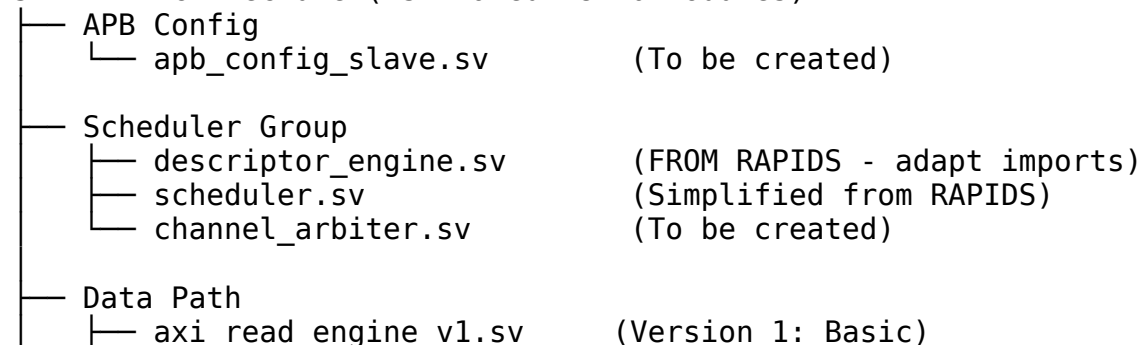
All 8 channels share: 1. Descriptor fetch AXI master 2. Data read AXI master 3. Data write AXI master 4. SRAM buffer 5. MonBus reporter

Arbitration is required! - Never assume exclusive access - All shared resources need arbiter logic - Priority-based round-robin (descriptor priority field)

22.4 Architecture Quick Reference

22.4.1 Block Organization

STREAM Architecture (Estimated: 8-10 modules)



—	axi_read_engine_v2.sv	(Version 2: Pipelined)
—	axi_write_engine_v1.sv	(Version 1: Basic)
—	axi_write_engine_v2.sv	(Version 2: Pipelined)
—	simple_sram.sv	(FROM RAPIDS - no changes)
—	MonBus Reporter	
—	monbus_reporter.sv	(To be created)

22.4.2 Module Status

Module	Source	Status	Notes
descriptor_engine.sv	RAPIDS (copy)	✓ Copied	Adapt #include only
simple_sram.sv	RAPIDS (copy)	✓ Copied	No changes needed
stream_pkg.sv	New	✓ Created	Descriptor format defined
stream_imports.svh	New	✓ Created	Package imports
scheduler.sv	RAPIDS (simplify)	⌚ Pending	Remove credit mgmt, control engines
apb_config_slave.sv	New	⌚ Pending	8 channel registers
axi_read_engine_v1.sv	New	⌚ Pending	Basic version
axi_write_engine_v1.sv	New	⌚ Pending	Basic version
channel_arbiter.sv	New	⌚ Pending	Priority-based round-robin
monbus_reporter.sv	New	⌚ Pending	STREAM event codes
stream_top.sv	New	⌚ Pending	Top-level integration

22.5 Common User Questions and Responses

22.5.1 Q: “How is STREAM different from RAPIDS?”

A: STREAM is intentionally simplified for tutorial purposes:

Simplifications: | Feature | RAPIDS | STREAM | |———|———|———| | **Interfaces** | APB + AXI + Network | APB + AXI only | | **Address Alignment** | Complex fixup | Aligned only | | **Credit Management** | Exponential encoding | Simple limits | | **Descriptor Length** | Chunks (4-byte) | Beats (data width) | | **Control Engines** | Control read/write | None (direct APB) |

Learning Path: 1. **STREAM** - Basic DMA with scatter-gather 2. **STREAM Extended** - Add alignment fixup 3. **RAPIDS** - Full complexity with network + credit management

 **See:** PRD.md Section 2.1 for complete comparison table

22.5.2 Q: “How do I kick off a transfer?”

A: Single APB write starts descriptor chain:

```
// Software writes descriptor address to channel register  
write_apb(ADDR_CH0_CTRL, 0x1000_0000); // Start address of descriptor
```

```
// STREAM automatically:  
// 1. Fetches descriptor from 0x1000_0000  
// 2. Parses src_addr, dst_addr, length  
// 3. Reads source data → SRAM  
// 4. Writes SRAM → destination  
// 5. Checks next_descriptor_ptr  
// - If != 0: Fetch next descriptor, repeat  
// - If == 0 or last flag set: Complete
```

Chained Descriptors:

```
Descriptor 0 @ 0x1000_0000:  
  src_addr = 0x2000_0000  
  dst_addr = 0x3000_0000  
  length = 64 beats  
  next_descriptor_ptr = 0x1000_0100 ← Points to next descriptor
```

```
Descriptor 1 @ 0x1000_0100:  
  src_addr = 0x2000_1000  
  dst_addr = 0x3000_1000  
  length = 32 beats  
  next_descriptor_ptr = 0x0000_0000 ← Last descriptor (0 = stop)
```

 **See:** PRD.md Section 3.2 for complete data flow

22.5.3 Q: “How many channels can I use?”

A: Maximum 8 independent channels:

Channel Operation: - Each channel has own FSM state - Each channel can have separate descriptor chain - All channels share: AXI masters, SRAM, MonBus

Arbitration: - Priority-based (descriptor priority field) - Round-robin within same priority - Prevents starvation with timeout

Example:

```
// Kick off 3 channels concurrently
write_apb(ADDR_CH0_CTRL, desc0_addr); // Channel 0: Priority 7
write_apb(ADDR_CH1_CTRL, desc1_addr); // Channel 1: Priority 5
write_apb(ADDR_CH2_CTRL, desc2_addr); // Channel 2: Priority 5

// Service order: CH0 → CH1 → CH2 → CH0 → CH1 → CH2 ...
```

 **See:** PRD.md Section 7 for arbitration details

22.5.4 Q: “What’s the descriptor format?”

A: 256-bit descriptor (4 × 64-bit words):

```
typedef struct packed {
    logic [63:0] reserved;           // [255:192] Reserved
    logic [7:0] priority;             // [207:200] Priority
    logic [3:0] channel_id;          // [199:196] Channel ID
    logic error;                     // [195] Error flag
    logic last;                       // [194] Last in chain
    logic interrupt;                 // [193] Generate interrupt
    logic valid;                     // [192] Valid descriptor
    logic [31:0] next_descriptor_ptr; // [191:160] Next descriptor
    address
    logic [31:0] length;              // [159:128] Length in BEATS
    ★
    logic [63:0] dst_addr;            // [127:64] Destination
    address
    logic [63:0] src_addr;            // [63:0] Source address
} descriptor_t;
```

★ **CRITICAL:** length is in **BEATS**, not bytes or chunks!

Example:

Data width = 512 bits = 64 bytes

Length = 16 beats

Total transfer = 16 × 64 = 1024 bytes

 **See:** PRD.md Section 4.2 for complete descriptor specification

22.5.5 Q: “How do I run STREAM tests?”

A: Multi-layered test approach (same as RAPIDS):

```
# 1. FUB (Functional Unit Block) Tests - Individual blocks
pytest projects/components/stream/dv/tests/fub_tests/scheduler/ -v
pytest
projects/components/stream/dv/tests/fub_tests/descriptor_engine/ -v
pytest projects/components/stream/dv/tests/fub_tests/ -v # All FUB
tests
```

```
# 2. Integration Tests - Multi-block scenarios
pytest projects/components/stream/dv/tests/integration_tests/ -v
```

```
# Run with waveforms for debugging
pytest projects/components/stream/dv/tests/fub_tests/scheduler/ --
vcd=debug.vcd
gtkwave debug.vcd
```

Test Organization: - **FUB tests:** Focus on individual block functionality - **Integration tests:** Verify block-to-block interfaces and complete data flows

22.6 Integration Patterns

22.6.1 Pattern 1: Basic STREAM Instantiation

```
stream_top #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .ADDR_WIDTH(64),
    .SRAM_DEPTH(4096)
) u_stream (
    // Clock and Reset
    .aclk                (system_clk),
    .aresetn              (system_rst_n),

    // APB Configuration Interface
    .s_apb_paddr          (apb_paddr),
    .s_apb_psel           (apb_psel),
    .s_apb_penable        (apb_penable),
    .s_apb_pwrite         (apb_pwrite),
    .s_apb_pwdata         (apb_pwdata),
    .s_apb_pready         (apb_pready),
    .s_apb_prdata         (apb_prdata),
```

```

.s_apb_pslverr      (apb_pslverr),

// AXI Master - Descriptor Fetch (256-bit)
.m_axi_desc_araddr   (desc_araddr),
.m_axi_desc_arlen    (desc_arlen),
.m_axi_desc_arsize   (desc_arsize),
.m_axi_desc_arvalid  (desc_arvalid),
.m_axi_desc_arready  (desc_arready),
.m_axi_desc_rdata    (desc_rdata),
.m_axi_desc_rresp    (desc_rresp),
.m_axi_desc_rlast    (desc_rlast),
.m_axi_desc_rvalid   (desc_rvalid),
.m_axi_desc_rready   (desc_rready),

// AXI Master - Data Read (parameterizable width)
.m_axi_rd_araddr     (rd_araddr),
// ... (full AXI4 AR + R channels)

// AXI Master - Data Write (parameterizable width)
.m_axi_wr_awaddr     (wr_awaddr),
// ... (full AXI4 AW + W + B channels)

// MonBus Output
.monbus_pkt_valid    (stream_mon_valid),
.monbus_pkt_ready    (stream_mon_ready),
.monbus_pkt_data     (stream_mon_data)
);

```

22.6.2 Pattern 2: Descriptor Creation (Software Model)

```

// C/C++ software model for descriptor creation
typedef struct {
    uint64_t src_addr;           // Source address (must be aligned!)
    uint64_t dst_addr;           // Destination address (must be
aligned!)
    uint32_t length;             // Transfer length in BEATS
    uint32_t next_descriptor_ptr; // Next descriptor address (0 =
last)
    uint8_t control;             // valid | interrupt | last | error |
channel_id | priority
} stream_descriptor_t;

// Create descriptor chain
stream_descriptor_t desc[2];

// Descriptor 0
desc[0].src_addr = 0x80000000ULL; // Aligned to 64B (512-bit data)

```

```

desc[0].dst_addr = 0x90000000ULL;
desc[0].length = 64; // 64 beats × 64 bytes = 4KB transfer
desc[0].next_descriptor_ptr = (uint32_t)&desc[1]; // Chain to next
desc[0].control = 0x01; // valid = 1

// Descriptor 1 (last)
desc[1].src_addr = 0x80001000ULL;
desc[1].dst_addr = 0x90001000ULL;
desc[1].length = 32; // 32 beats × 64 bytes = 2KB transfer
desc[1].next_descriptor_ptr = 0; // Last descriptor
desc[1].control = 0x45; // valid | last | interrupt

// Kick off transfer
write_apb_reg(CH0_CTRL, (uint32_t)&desc[0]);

```

22.6.3 Pattern 3: MonBus Integration

```

// Always add downstream FIFO for MonBus
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_stream_mon_fifo (
    .i_clk      (aclk),
    .i_rst_n    (aresetn),
    .i_data     (monbus_pkt_data),
    .i_valid    (monbus_pkt_valid),
    .o_ready    (monbus_pkt_ready),
    .o_data     (fifo_mon_data),
    .o_valid    (fifo_mon_valid),
    .i_ready    (consumer_ready)
);

```

22.7 Anti-Patterns to Catch

22.7.1 X Anti-Pattern 1: Adding Alignment Fixup

x WRONG:

"Let me add alignment fixup logic to handle unaligned addresses..."

✓ CORRECTED:

"STREAM intentionally requires aligned addresses for tutorial simplicity."

If you need unaligned transfers, that's a great learning exercise **for** extending STREAM, **or use** RAPIDS which has full alignment support."

22.7.2 X Anti-Pattern 2: Using Length in Bytes

x WRONG:

```
descriptor.length = 4096; // Thinking this is 4096 bytes
```

✓ CORRECTED:

```
// Length is in BEATS, not bytes!  
// For 512-bit data width (64 bytes per beat):  
descriptor.length = 4096 / 64; // = 64 beats for 4096 bytes
```

22.7.3 X Anti-Pattern 3: Circular Buffer Descriptors

x WRONG:

```
// Descriptor chain that loops back  
desc[9].next_descriptor_ptr = &desc[0]; // Circular!
```

✓ CORRECTED:

```
"STREAM does not support circular buffers (no stop condition).  
Always terminate chains explicitly:  
desc[last].next_descriptor_ptr = 0; // Stop  
desc[last].last = 1; // Explicit last flag
```

22.7.4 X Anti-Pattern 4: Assuming Exclusive Channel Access

x WRONG:

```
// Assume channel has exclusive AXI master access  
assign m_axi_arvalid = channel_request; // No arbitration!
```

✓ CORRECTED:

```
// All channels share AXI masters - arbitration required  
channel_arbiter u_arbiter (  
    .ch_requests (channel_requests[7:0]),  
    .ch_grant     (channel_grant[7:0]),  
    .axi_master_available (axi_master_idle)  
);
```

22.8 Debugging Workflow

22.8.1 Issue: Descriptor Not Fetched

Check in order: 1. ✓ APB write to CHx_CTRL register successful? 2. ✓ Descriptor address valid? 3. ✓ AXI descriptor master not stalled? 4. ✓ Descriptor memory accessible? 5. ✓ Arbiter granting descriptor fetch?

Debug commands:

```

pytest projects/components/stream/dv/tests/fub_tests/scheduler/ -v -s
# Verbose test
pytest projects/components/stream/dv/tests/fub_tests/scheduler/ --
vcd=debug.vcd
gtkwave debug.vcd # Inspect FSM state transitions

```

22.8.2 Issue: Data Transfer Stalls

Check in order: 1. ✓ SRAM buffer depth sufficient? 2. ✓ Source/destination addresses aligned? 3. ✓ AXI read/write engines getting grants? 4. ✓ Downstream AXI ready signals asserted? 5. ✓ Channel not in error state?

Waveform Analysis: - Check SRAM read/write pointers - Verify AXI AR/AW/R/W/B handshakes - Inspect scheduler FSM state - Check arbiter grant signals

22.8.3 Issue: Chained Descriptors Not Following

Check in order: 1. ✓ next_descriptor_ptr != 0? 2. ✓ last flag not set prematurely? 3. ✓ Descriptor fetch completing successfully? 4. ✓ Scheduler transitioning to CHAIN_CHECK state? 5. ✓ MonBus showing CHAIN_FETCH event?

22.9 Testing Guidance

22.9.1 Test Organization

```

projects/components/stream/dv/tests/
├── fub_tests/                # Individual block tests
│   ├── descriptor_engine/    # Adapt from RAPIDS tests
│   ├── scheduler/           # Simplified scheduler tests
│   ├── axi_engines/         # Read/write engine tests
│   └── sram/                # SRAM tests (from RAPIDS)
├── integration_tests/       # Multi-block scenarios
│   ├── single_channel/      # Single channel transfers
│   ├── multi_channel/       # 8-channel concurrent
│   ├── chained_descriptors/ # Descriptor chain tests
│   └── error_handling/      # Error recovery tests

```

22.9.2 Test Levels

Basic (Quick Smoke Tests): - Single descriptor, single channel - Aligned addresses, simple transfers - ~30 seconds runtime

Medium (Typical Scenarios): - Multiple descriptors, 2-4 channels - Chained descriptors (2-3 deep) - ~90 seconds runtime

Full (Comprehensive Validation): - All 8 channels concurrent - Long descriptor chains (10+ deep) - Error injection, stress testing - ~180 seconds runtime

22.10 Key Documentation Links

22.10.1 Always Reference These

This Subsystem: - projects/components/stream/PRD.md - **Complete specification** - projects/components/stream/README.md - Quick start guide (to be created) - projects/components/stream/known_issues/ - Bug tracking

Related: - projects/components/rapids/PRD.md - Parent architecture (for comparison) - rtl/amba/PRD.md - MonBus integration - /PRD.md - Master requirements - /CLAUDE.md - Repository guide

22.11 Quick Commands

View complete specification

```
cat projects/components/stream/PRD.md
```

Check package definition

```
cat projects/components/stream/rtl/includes/stream_pkg.sv
```

Run tests (once created)

```
pytest projects/components/stream/dv/tests/fub_tests/ -v
```

```
pytest projects/components/stream/dv/tests/integration_tests/ -v
```

Lint (once RTL created)

```
verilator --lint-only
```

```
projects/components/stream/rtl/stream_fub/scheduler.sv
```

Search for modules

```
find projects/components/stream/rtl/ -name "*.sv" -exec grep -H  
"^module" {} \;
```

22.12 Remember

1.  **Tutorial focus** - Intentional simplifications for learning
2.  **Reuse from RAPIDS** - Descriptor engine, SRAM, patterns

3.  **Length in beats** - Not bytes or chunks!
 4.  **Aligned addresses** - No fixup logic
 5.  **Chained descriptors** - No circular buffers
 6.  **8 channels** - Shared resources, arbitration required
 7.  **MonBus standard** - Same format as AMBA/RAPIDS
 8.  **Testbench reuse** - Always create TB classes in bin/TBClasses/stream/
-

22.13 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

/mnt/data/github/rtldesignsherpa/projects/components/stream/docs/

Quick Command: Use the provided shell script:

```
cd /mnt/data/github/rtldesignsherpa/projects/components/stream/docs
./generate_pdf.sh
```

The shell script will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the stream_spec index file 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

 **See:** bin/md_to_docx.py for complete implementation details

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